

Stochastic Volatility Selection in Bayesian Vector Autoregressions

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ABSTRACT

This paper develops a Bayesian vector autoregression with stochastic-volatility variable selection (BVAR-SVVS) that determines, variable by variable, whether each series is better characterized by a stochastic-volatility or constant-volatility specification. The model uses a spike-and-slab prior on the innovation variance of each variable's log-volatility process within a hierarchical Minnesota prior framework. This approach avoids the combinatorial burden of explicit model comparison while remaining computationally feasible in large systems. Using U.S. quarterly macroeconomic datasets of increasing dimension, we find marked heterogeneity in volatility dynamics across variables. Real activity variables are systematically assigned to stochastic-volatility specifications, whereas financial and price-related series more often favor constant volatility. Relative to Bayesian VARs that impose either stochastic volatility or constant volatility uniformly across all variables, the proposed approach delivers sizable gains in density forecasting. The improvements are strongest in large systems and at medium-to-long forecast horizons, suggesting that parsimony in the second moment can improve predictive accuracy without sacrificing flexibility where time-varying volatility is empirically warranted.

KEYWORDS

Spike-and-slab prior; density forecasting; Minnesota prior; macroeconomic forecasting

JEL CLASSIFICATION

C11; C32; C53; E37

1. Introduction

This paper develops a Bayesian VAR with stochastic-volatility variable selection (BVAR-SVVS). The model classifies, variable by variable, whether volatility is better represented as stochastic or constant. Bayesian VARs with stochastic volatility have become a standard framework for joint density forecasting in macroeconomics (Clark, 2011; Primiceri, 2005; Cogley and Sargent, 2005). Yet they typically impose the volatility specification uniformly across the system, assuming that every variable follows a stochastic-volatility process regardless of empirical evidence. In large systems, this restriction can be unnecessarily strong. Some series exhibit persistent, forecast-relevant volatility variation, whereas others show little evidence of such dynamics. BVAR-SVVS relaxes this restriction by allowing the volatility specification to differ across variables.

For density forecasting, not all volatility variation is equally informative (Jurado et al., 2015).¹ Persistent and economically large shifts in predictive uncertainty, such as those surrounding the Global Financial Crisis or the COVID-19 pandemic, warrant a stochastic-volatility specification. Transient, quickly mean-reverting fluctuations do not. For financial variables such as equity returns, volatility clustering that is prominent at daily frequencies can be substantially attenuated by quarterly temporal aggregation (Drost and Nijman, 1993). Forcing all variables into an SV process can therefore introduce more latent variation than the data can support (Carriero et al., 2016). When the true conditional variance is close to constant, a persistent log-volatility process adds latent state uncertainty that accumulates with the forecast horizon. This can inflate predictive densities and is most costly at medium-to-long horizons.

BVAR-SVVS extends variable selection to the variance equation, the dimension along which the BVAR-SV literature has been most permissive.² Following George

¹See also Rossi and Sekhposyan (2015) for a related discussion of forecast-error-based uncertainty measures.

²Variable selection in the mean equation of Bayesian VARs is well-studied. See Korobilis (2013) for an early application and Huber and Feldkircher (2019) for an adaptive shrinkage variant.

and McCulloch (1997), we place a spike-and-slab prior on the innovation variance of each variable’s log-volatility process. The slab captures variables with persistent and quantitatively relevant volatility variation, assigning them a stochastic-volatility process. The spike captures variables with negligible volatility dynamics, assigning them a constant-volatility process. Enumerating all 2^n specifications is infeasible as n grows. Our formulation circumvents this problem by adding two Gibbs blocks to a standard BVAR-SV sampler. It builds on the equation-by-equation representation of Carriero et al. (2019), which keeps estimation tractable in large systems.

It should be noted that the selection mechanism is applied to the orthogonalized shocks obtained from the equation-by-equation triangular representation, rather than to the reduced-form residuals directly. Because the spike-and-slab prior is placed at this equation level, it does not depend on the particular way the covariance matrix is parameterized elsewhere in the model. This separability means the same selection prior can be grafted onto alternative volatility specifications without modification. For instance, in a factor stochastic-volatility model, the identical spike-and-slab structure can be applied to the idiosyncratic volatility process of each factor equation.

We evaluate the prediction performance of the BVAR-SVVS using a quarterly U.S. macroeconomic and financial dataset spanning 1978Q1 to 2024Q3. Our out-of-sample forecasting experiment considers systems of 3, 10 and 25 variables. The estimated volatility structure is notably heterogeneous. Real activity variables, such as output, consumption, and industrial production, tend to be classified as stochastic-volatility processes. By contrast, several financial and price-related series more often favor constant volatility under the baseline calibration.

Relative to the BVAR-SV benchmark, BVAR-SVVS improves the joint average log predictive score in the largest system at medium-to-long horizons, where redundant time-varying volatility is most costly. Relative to the BVAR with constant volatil-

ity (BVAR-CV) benchmark, the gains are broader across system sizes, although they weaken at the longest horizon in the largest system. A decomposition of these gains suggests that BVAR-SVVS captures major volatility episodes, including the Global Financial Crisis and the COVID-19 pandemic. At the same time, it avoids some redundant volatility variation in calmer periods.

The remainder of the paper proceeds as follows. Section 2 introduces the benchmark BVAR-SV and BVAR-CV models. Section 3 develops the volatility-specification selection prior. Section 4 details the Bayesian estimation procedure. Section 5 presents the empirical forecasting results. Section 6 examines robustness to estimated contemporaneous dependence. Section 7 concludes.

2. Benchmark BVARs: BVAR-SV and BVAR-CV

Both benchmark specifications share the same conditional mean but differ in their conditional covariance structure:

$$\text{BVAR-CV: } y_t \mid \mathcal{I}_{t-1}, \theta, \Omega \sim \mathcal{N}(\Phi_0 + \Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p}, \Omega), \quad (2.1)$$

$$\text{BVAR-SV: } y_t \mid \mathcal{I}_{t-1}, \theta, h_t \sim \mathcal{N}(\Phi_0 + \Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p}, \Omega_t), \quad (2.2)$$

where $y_t = (y_{1,t}, \dots, y_{n,t})'$ is the $n \times 1$ vector of endogenous variables, $\Phi_0 \in \mathbb{R}^n$ is the intercept, $\Phi_j \in \mathbb{R}^{n \times n}$ for $j = 1, \dots, p$ are the lag coefficient matrices collected in θ , and $\mathcal{I}_{t-1}$ denotes the information set at $t-1$. The vector $h_t = (h_{1,t}, \dots, h_{n,t})'$ collects the log-volatilities introduced below.

Under BVAR-SV, the time-varying covariance factorizes as $\Omega_t = B_0^{-1} \Sigma_t (B_0')^{-1}$, with B_0 lower triangular with unit diagonal and off-diagonal elements b_{ij} ($1 \leq j < i \leq n$), and $\Sigma_t = \text{diag}(e^{h_{1,t}}, \dots, e^{h_{n,t}})$ collecting the variable-specific volatilities. The reason why we consider the Cholesky specification for the empirical illustration is that it

outperforms more restrictive alternatives, such as the common stochastic volatility model, in large systems according to the study of Chan (2023).³

Each log-volatility follows the AR(1) process

$$h_{i,t} \mid h_{i,t-1}, \alpha_i, \phi_i, \sigma_{i,h}^2 \sim \mathcal{N}(\alpha_i + \phi_i h_{i,t-1}, \sigma_{i,h}^2), \quad i = 1, \dots, n, \quad (2.3)$$

with level α_i , persistence ϕ_i , and innovation variance $\sigma_{i,h}^2$, as in Cogley and Sargent (2005) and Primiceri (2005).

For posterior computation, we use the equation-by-equation representation of Carriero et al. (2019). This representation factorizes the joint Gaussian likelihood into n conditionally univariate regressions, making posterior simulation tractable in large systems. Stacking observations over $t = 1, \dots, T$, let $y_i = (y_{i,1}, \dots, y_{i,T})'$ denote the $T \times 1$ vector of observations for variable i . The i -th triangular equation can be written as

$$y_i \mid X_i, \vartheta_i, H_i \sim \mathcal{N}(X_i \vartheta_i, H_i), \quad i = 1, \dots, n, \quad (2.4)$$

where $X_i = (Z, -Y_{1:i-1})$ combines the lagged regressors Z (with t -th row $(1, y'_{t-1}, \dots, y'_{t-p})$) and the contemporaneous observations $Y_{1:i-1}$ of variables ordered before i . The coefficient vector

$$\vartheta_i = (\beta_{0,i}, \beta_{1,i,1}, \dots, \beta_{1,i,n}, \dots, \beta_{p,i,1}, \dots, \beta_{p,i,n}, b_{i,1}, \dots, b_{i,i-1})' \in \mathbb{R}^{np+i} \quad (2.5)$$

stacks the intercept $\beta_{0,i}$, lag coefficients $\beta_{l,i,j}$ for $l = 1, \dots, p$ and $j = 1, \dots, n$, and contemporaneous coefficients $b_{i,1}, \dots, b_{i,i-1}$. The error covariance $H_i =$

³The triangular factorization further accommodates cross-variable shrinkage and flexible priors on B_0 , both of which are important for controlling overfitting as n grows.

$\text{diag}(e^{h_{i,1}}, \dots, e^{h_{i,T}})$ reduces to $\omega_i^2 I_T$ under BVAR-CV.

In the baseline specification, the off-diagonal elements of B_0 are heavily shrunk toward zero. This isolates the role of volatility selection from cross-equation dependence in the contemporaneous covariance, so that any subsequent comparison reflects volatility dynamics rather than additional flexibility in the covariance structure. For robustness, in Section 6, we relax this restriction and allow the contemporaneous-coefficient prior scale to be estimated from the data.

We impose a Minnesota-type prior on the coefficients in ϑ_i . For the lag coefficients ($l \geq 1$),

$$\beta_{l,i,j} \mid \lambda_1, \lambda_2 \sim \mathcal{N}(m_{l,i,j}, v_{l,i,j}), \quad (2.6)$$

with prior mean $m_{l,i,j} = 0.5$ for the first own lag ($l = 1, i = j$) and zero otherwise,⁴ and prior variance

$$v_{l,i,j} = \begin{cases} \frac{\lambda_1}{l^2}, & i = j, \\ \frac{\lambda_1 \lambda_2 \hat{\omega}_i^2}{l^2 \hat{\omega}_j^2}, & i \neq j, \end{cases} \quad (2.7)$$

where $\hat{\omega}_i$ and $\hat{\omega}_j$ are the residual standard deviations from univariate AR(4) regressions (Bańbura et al., 2010). The intercept and contemporaneous coefficients receive independent Gaussian priors,

$$\beta_{0,i} \mid \lambda_3 \sim \mathcal{N}(0, \lambda_3) \text{ and } b_{i,j} \mid \lambda_4 \sim \mathcal{N}(0, \lambda_4) \text{ for } 1 \leq j < i \leq n. \quad (2.8)$$

⁴Since the variables are specified in stationary transformations, we do not impose a random-walk prior. The prior mean of 0.5 provides a mildly persistent benchmark while leaving the degree of persistence primarily determined by the data.

Among the four hyperparameters, λ_1 controls overall tightness, λ_2 governs cross-variable shrinkage, and λ_3 and λ_4 scale the intercept and contemporaneous-coefficient priors, respectively.

Rather than fixing λ_1 and λ_2 , we follow Giannone et al. (2015) and place Gamma hyperpriors on them such that

$$\lambda_1 \sim \mathcal{G}(e_{11}, e_{12}) \text{ and } \lambda_2 \sim \mathcal{G}(e_{21}, e_{22}), \quad (2.9)$$

where $\mathcal{G}(a, b)$ denotes a Gamma distribution with shape a and rate b , so that $\mathbb{E}[\lambda] = a/b$. These hierarchical priors allow the data to determine the degree of shrinkage and help maintain predictive efficiency across different system sizes. Section 6 extends the same hierarchical treatment to λ_3 and λ_4 .

3. BVARs with Stochastic Volatility Selection: BVAR-SVVS

BVAR-SVVS augments the BVAR-SV specification in equations (2.2)–(2.3) with a variable-specific binary indicator $\gamma_i \in \{0, 1\}$. The indicator classifies variable i as following either a stochastic-volatility process ($\gamma_i = 1$) or a constant-volatility process ($\gamma_i = 0$). Although we use the term *variable-specific* throughout for readability, the indicator formally classifies the volatility of each orthogonalized equation-level shock rather than the reduced-form variance of the corresponding observed variable. Because reduced-form variances aggregate shock-level volatilities through B_0^{-1} , a variable whose own shock is classified as CV may still display time-varying reduced-form variance when other equation-level shocks carry SV.

Following the spike-and-slab framework of George and McCulloch (1997), we place

a hierarchical prior on γ_i and the volatility-block parameters $(\alpha_i, \phi_i, \sigma_{i,h}^2)$:

$$\gamma_i \mid p \sim \text{Bernoulli}(p), \quad (3.1)$$

$$p \sim \text{Beta}(a_p, b_p), \quad (3.2)$$

$$\alpha_i \mid \gamma_i \sim \begin{cases} \mathcal{N}(\mu_\alpha^{SV}, \tau_\alpha^{SV}), & \gamma_i = 1, \\ \mathcal{N}(\mu_{\alpha,i}^{CV}, \tau_\alpha^{CV}), & \gamma_i = 0, \end{cases} \quad (3.3)$$

$$\phi_i \mid \gamma_i \sim \begin{cases} \mathcal{N}(\mu_\phi^{SV}, \tau_\phi^{SV}), & \gamma_i = 1, \\ \mathcal{N}(\mu_\phi^{CV}, \tau_\phi^{CV}), & \gamma_i = 0, \end{cases} \quad (3.4)$$

$$\text{and } \sigma_{i,h}^2 \mid \gamma_i \sim \begin{cases} \mathcal{IG}(\nu^{SV}, S^{SV}), & \gamma_i = 1, \\ \mathcal{IG}(\nu^{CV}, S^{CV}), & \gamma_i = 0, \end{cases} \quad (3.5)$$

where $\mathcal{IG}(\nu, S)$ denotes an Inverse-Gamma distribution with prior mean $S/(\nu - 1)$, and $\mu_{\alpha,i}^{CV}$ is variable-specific. We use $a_p = b_p = 1$, which gives a uniform prior for the system-level SV probability p and centers it at 0.5. This prior is symmetric between the SV and CV specifications, while allowing p to be updated jointly with the indicators $\{\gamma_i\}_{i=1}^n$.

The two components are designed to represent distinct volatility specifications. The slab component centers prior mass on persistent log-volatility dynamics, while the spike component concentrates prior mass near a constant-variance process. This separation allows the model to select the volatility specification variable by variable, rather than imposing stochastic volatility uniformly across the full system.

The hyperparameters are calibrated to create a sharp separation between the two components. For the slab ($\gamma_i = 1$), we set $\mu_\phi^{SV} = 0.99$, $\tau_\phi^{SV} = 0.01$, $\mu_\alpha^{SV} = 0$, $\tau_\alpha^{SV} = 0.01$, and $(\nu^{SV}, S^{SV}) = (10, 0.5)$. This prior places mass on persistent log-volatility

processes with moderate innovation variance.

For the spike ($\gamma_i = 0$), we set $\mu_\phi^{CV} = 0$, $\tau_\phi^{CV} = 10^{-7}$, $\tau_\alpha^{CV} = 10^{-7}$, $\mu_{\alpha,i}^{CV} = 2 \log \hat{\omega}_i$, and $S^{CV} = 10^{-7}$. The tight prior on ϕ_i (small τ_ϕ^{CV}) forces the log-volatility process to be non-persistent under the CV component, and $S^{CV} = 10^{-7}$ concentrates the prior on $\sigma_{i,h}^2$ near zero so that the innovation variance of $h_{i,t}$ is effectively shut down. The tight prior on α_i (small τ_α^{CV}) anchors the long-run log-variance at the AR(4) residual variance scale $2 \log \hat{\omega}_i$. According to our empirical experiment, without this anchoring, the CV-component posterior of α_i drifts upward across nearly all variables, raising the constant-variance baseline systematically above the BVAR-CV benchmark. The resulting predictive densities become overdispersed, eroding the parsimony gain of CV classification.

Since $\mu_{\alpha,i}^{CV} = 2 \log \hat{\omega}_i$, the implied conditional variance is centered at $e^{h_{i,t}} \approx \hat{\omega}_i^2$. The parameter ν^{CV} controls the concentration of the prior for $\sigma_{i,h}^2$ under the CV component. This concentration is central to the sensitivity of the SV/CV specification choice, and Section 5 discusses the calibration in detail. Consequently, these hyperparameter choices keep the CV component close to a non-persistent, constant-variance specification, while the SV component allows persistent volatility dynamics.

4. Bayesian Estimation

We sample from the joint posterior using a Gibbs sampler that cycles through the parameter blocks listed below. In the empirical implementation, we run 11,000 iterations and discard the first 1,000 as burn-in. Throughout, we suppress from each conditioning set parameters that do not enter the corresponding full conditional. In each iteration, the parameters, variable-specific binary indicators and SVs are simulated according to the following algorithm:

- (1) Draw $\vartheta_i \mid y_i, h_i, \lambda$, for $i = 1, \dots, n$.

- (2) Draw $s_i \mid y_i^*, h_i, \sigma_{i,h}^2$, for $i = 1, \dots, n$.
- (3) Draw $h_i \mid y_i^*, \alpha_i, \phi_i, \sigma_{i,h}^2, s_i$, for $i = 1, \dots, n$.
- (4) Draw $\lambda \mid \vartheta$.
- (5) Draw $(\alpha_i, \phi_i) \mid h_i, \gamma_i, \mu, \tau, \sigma_{i,h}^2$, for $i = 1, \dots, n$.
- (6) Draw $\sigma_{i,h}^2 \mid h_i, \alpha_i, \phi_i, \gamma_i, \nu, S$, for $i = 1, \dots, n$.
- (7) Draw $\gamma_i \mid \alpha_i, \phi_i, \sigma_{i,h}^2, \mu, \tau, \nu, S, p$, for $i = 1, \dots, n$.
- (8) Draw $p \mid \gamma$.

The vector s_i collects the mixture indicators used in the log-chi-squared approximation of Kim et al. (1998) and Omori et al. (2007), and y_i^* is the vector of log-squared residuals that drives the SV measurement equation. The collections μ and τ gather the SV and CV prior means and variances of (α_i, ϕ_i) , while $\nu = (\nu^{SV}, \nu^{CV})$ and $S = (S^{SV}, S^{CV})$ gather the inverse-Gamma hyperparameters of $\sigma_{i,h}^2$. All remaining symbols are defined in Sections 2 and 3.

All steps except Steps 7 and 8 follow standard procedures for Bayesian VARs with stochastic volatility, with derivations collected in Appendix C. Given that sampling p is straightforward, the remainder of this section focuses on Step 7, the sampling of the volatility-specification indicator γ_i .

4.1. Sampling the Volatility-Specification Indicator

Conditional on the current draws of $(\alpha_i, \phi_i, \sigma_{i,h}^2)$ and the prior inclusion probability p , the specification indicator γ_i is drawn from its Bernoulli full conditional distribution. By Bayes' rule,

$$\pi(\gamma_i \mid \alpha_i, \phi_i, \sigma_{i,h}^2, p) \propto \pi(\alpha_i, \phi_i, \sigma_{i,h}^2 \mid \gamma_i) \pi(\gamma_i \mid p). \quad (4.1)$$

To evaluate the relative support for the SV and CV components, define

$$R_i^{SV} = \mathcal{N}(\alpha_i \mid \mu_\alpha^{SV}, \tau_\alpha^{SV}) \mathcal{N}(\phi_i \mid \mu_\phi^{SV}, \tau_\phi^{SV}) \mathcal{IG}(\sigma_{i,h}^2 \mid \nu^{SV}, S^{SV}), \quad (4.2)$$

$$R_i^{CV} = \mathcal{N}(\alpha_i \mid \mu_{\alpha,i}^{CV}, \tau_\alpha^{CV}) \mathcal{N}(\phi_i \mid \mu_\phi^{CV}, \tau_\phi^{CV}) \mathcal{IG}(\sigma_{i,h}^2 \mid \nu^{CV}, S^{CV}). \quad (4.3)$$

The posterior probability that variable i follows a stochastic-volatility process is then

$$\Pr(\gamma_i = 1 \mid \alpha_i, \phi_i, \sigma_{i,h}^2, p) = \frac{R_i^{SV} p}{R_i^{SV} p + R_i^{CV} (1 - p)}. \quad (4.4)$$

The indicator is subsequently drawn as

$$\gamma_i \sim \text{Bernoulli} \left(\frac{R_i^{SV} p}{R_i^{SV} p + R_i^{CV} (1 - p)} \right). \quad (4.5)$$

This step compares the support assigned by the slab and spike components at the current draw of the volatility parameters. A larger value of this conditional allocation probability favors assignment to the SV component at the current draw.

Importantly, the classification of γ_i is determined jointly by the full volatility-parameter block $(\alpha_i, \phi_i, \sigma_{i,h}^2)$. Posterior support for SV therefore does not depend solely on the persistence parameter ϕ_i , but also reflects the intercept and innovation variance of the log-volatility process. In the implementation, we initialize $\gamma_i = 0$ for all i , corresponding to the parsimonious CV specification, so that variables move to the SV component only when supported by the data.

Two aspects of this selection step deserve clarification. First, equation (4.4) compares the prior density of the current draw $(\alpha_i, \phi_i, \sigma_{i,h}^2)$ under the SV and CV components, not a marginal or predictive likelihood. This is a conditional allocation update for the latent mixture indicator γ_i , given the current draw of the volatility parame-

ters. Although the SV and CV regions are defined by the prior calibration, the allocation is not mechanically prior-fixed. The data enter through the conditional draws of $(\alpha_i, \phi_i, \sigma_{i,h}^2)$, and γ_i is updated by evaluating where the current volatility-parameter draw receives greater support. Second, the tight CV priors ($\tau_\phi^{CV} = 10^{-7}$, $\tau_\alpha^{CV} = 10^{-7}$) are deliberate. Without a sufficiently tight spike, the SV and CV components overlap substantially in parameter space, and γ_i cannot meaningfully distinguish them. The sharpness of the spike is what operationalizes the binary classification; relaxing it collapses the distinction between components and effectively reduces the procedure to unconstrained SV estimation.

5. Empirical Results

5.1. Data and Forecast Design

The empirical analysis uses a quarterly U.S. macroeconomic and financial dataset spanning 1978Q1 to 2024Q3, constructed following Crump et al. (2025). The resulting dataset contains 25 variables covering real activity, prices, labor markets, interest rates, financial markets, and commodity prices. We exclude the federal funds rate because its behavior at the zero lower bound renders the series uninformative as a measure of the conventional monetary policy stance over much of our sample. Series originally recorded at higher frequencies are aggregated to quarterly averages; most enter the model as quarter-over-quarter log growth rates, while stationary series enter in levels. Appendix Table A1 provides variable definitions, transformations, and model inclusion sets for each system.

We estimate all models on three nested systems of increasing dimension: a small system of three variables (real GDP growth, core PCE inflation, and the 2-year Treasury yield), a medium system of 10 variables following Wright (2013), and a large system containing all 25 variables. This design allows us to assess both how the BVAR-SVVS

volatility classification responds to the size of the information set and how forecast gains from stochastic volatility selection scale with model dimension. Forecasts are produced recursively using an expanding estimation window. The initial estimation period runs from 1978Q1 to 2000Q4, and the out-of-sample evaluation period spans 2002Q1 to 2024Q3, yielding T_h forecasts at each horizon h .

Point forecast accuracy is evaluated using the root mean squared forecast error (RMSFE). For variable i , horizon h , and model $m \in \{\text{BVAR-SV}, \text{BVAR-CV}\}$,

$$\text{RMSFE}_{i,h}^m = \left[\frac{1}{T_h} \sum_{t=t_0}^{T-h} \left(y_{i,t+h} - \hat{y}_{i,t+h|t}^m \right)^2 \right]^{1/2}, \quad (5.1)$$

where $\hat{y}_{i,t+h|t}^m$ denotes the posterior mean forecast from model m given information $\mathcal{I}_t$, t_0 is the first forecast origin, and $T_h = T - h - t_0 + 1$ is the number of evaluation periods at horizon h . To assess relative performance, we report

$$\text{RelRMSFE}_{i,h}^m = \frac{\text{RMSFE}_{i,h}^{\text{BVAR-SVVS}}}{\text{RMSFE}_{i,h}^m}, \quad (5.2)$$

so that $\text{RelRMSFE}_{i,h}^m < 1$ indicates that BVAR-SVVS outperforms benchmark m .

Density forecast accuracy is evaluated using the joint average log predictive score (joint ALPS),

$$\text{joint ALPS}_h^m = \frac{1}{T_h} \sum_{t=t_0}^{T-h} \log p^m(y_{t+h} \mid \mathcal{I}_t), \quad (5.3)$$

where y_{t+h} is the realized $n \times 1$ outcome vector and $p^m(\cdot \mid \mathcal{I}_t)$ is the joint predictive density of model m , approximated from the MCMC draws via Monte Carlo integration.

The differential

$$\Delta_{\text{joint ALPS}}^m = \text{joint ALPS}_h^{\text{BVAR-SVVS}} - \text{joint ALPS}_h^m, \quad (5.4)$$

measures the density-forecast gain of BVAR-SVVS relative to benchmark m , with $\Delta_{\text{joint ALPS}}^m > 0$ indicating that BVAR-SVVS assigns higher average probability to realized outcomes.

5.2. Variable-Level Classification

The variable-level classification depends on the hyperparameters governing the volatility-selection prior. We use ν^{CV} as the calibration parameter because it adjusts the restrictiveness of the CV benchmark without changing the defining features of the two components. Changing the persistence means or variances would blur this distinction, allowing either noise-like volatility under the SV component or weakly time-varying volatility under the CV component. Similarly, the tight CV prior on α_i anchors the constant-variance component to the variable-specific scale implied by the AR(4) residual variance.

We also keep ν^{SV} fixed at the value used for the benchmark BVAR-SV model. This preserves the interpretation of the slab as the same stochastic-volatility specification used in the BVAR-SV benchmark. Thus, varying ν^{CV} changes the classification threshold by controlling how much log-volatility innovation variance is tolerated under the CV component, without redefining the SV component itself. The resulting path moves from CV to SV classifications while leaving the shapes of the two components intact.

Table 1.: Classification of Constant and Stochastic Variables by ν_{CV}

ν_{CV}	5.00	10.00	15.00	20.00	25.00	30.00	35.00	40.00	50.00	60.00	70.00	80.00	90.00	100.00
GDP														
CONS														
INV														
EXP														
IMP														
GOV														
DISP														
GDPP														
PCE														
PCEX														
CPI														
POP														
UEMP														
WAGE														
HOUS														
IPI														
CAP														
SENT														
TR2														
TR5														
TR10														
AAA														
BAA														
SP500														
WTI														
# CV	23.00	17.00	13.00	12.00	10.00	9.00	8.00	6.00	5.00	3.00	2.00	2.00	1.00	1.00
Joint ALPS ($h = 1$)	-65.98	-73.76	-73.95	-73.13	-73.09	-72.26	-71.81	-71.47	-52.04	-55.24	-57.44	-57.44	-54.94	-56.39
Avg. RMSFE ($h = 1$)	3.05	3.10	3.10	3.10	3.10	3.07	3.05	3.04	3.04	3.04	3.03	3.03	3.02	3.03
Joint ALPS ($h = 4$)	-100.17	-103.49	-99.55	-101.45	-110.58	-111.39	-115.48	-112.10	-105.45	-100.88	-104.02	-101.59	-102.64	-108.78
Avg. RMSFE ($h = 4$)	66.38	3.36	3.36	3.35	3.35	3.35	3.35	3.35	3.37	3.38	3.38	3.36	3.36	3.36
Joint ALPS ($h = 8$)	-105.25	-114.73	-119.90	-122.20	-128.18	-127.26	-129.50	-135.78	-150.81	-164.06	-144.95	-146.73	-158.11	-164.88
Avg. RMSFE ($h = 8$)	93.50	3.70	3.69	3.67	3.67	3.68	3.67	3.68	3.73	3.73	3.74	3.73	3.73	3.72
Joint ALPS ($h = 16$)	-105.08	-116.43	-126.35	-117.82	-135.51	-130.48	-137.38	-136.45	-126.87	-133.96	-140.21	-149.24	-136.40	-144.85
Avg. RMSFE ($h = 16$)	48.48	3.92	3.93	3.90	3.91	3.91	3.90	3.91	4.04	4.07	4.06	4.06	4.05	4.05

Note: Shaded cells denote CV specifications, while blank cells correspond to SV specifications. The classifications are based on full-sample estimation and summarize how the mixture composition varies with ν_{CV} . In the rolling forecasting exercise, the selected classification may change over time as the information set expands; see Figure 1. The first row report the number of variables classified as CV and the corresponding joint ALPS values across forecast horizons $h = \{1, 4, 8, 16\}$. The RMSEF values reported in parentheses are the cross-variable averages of RMSEF at each horizon.

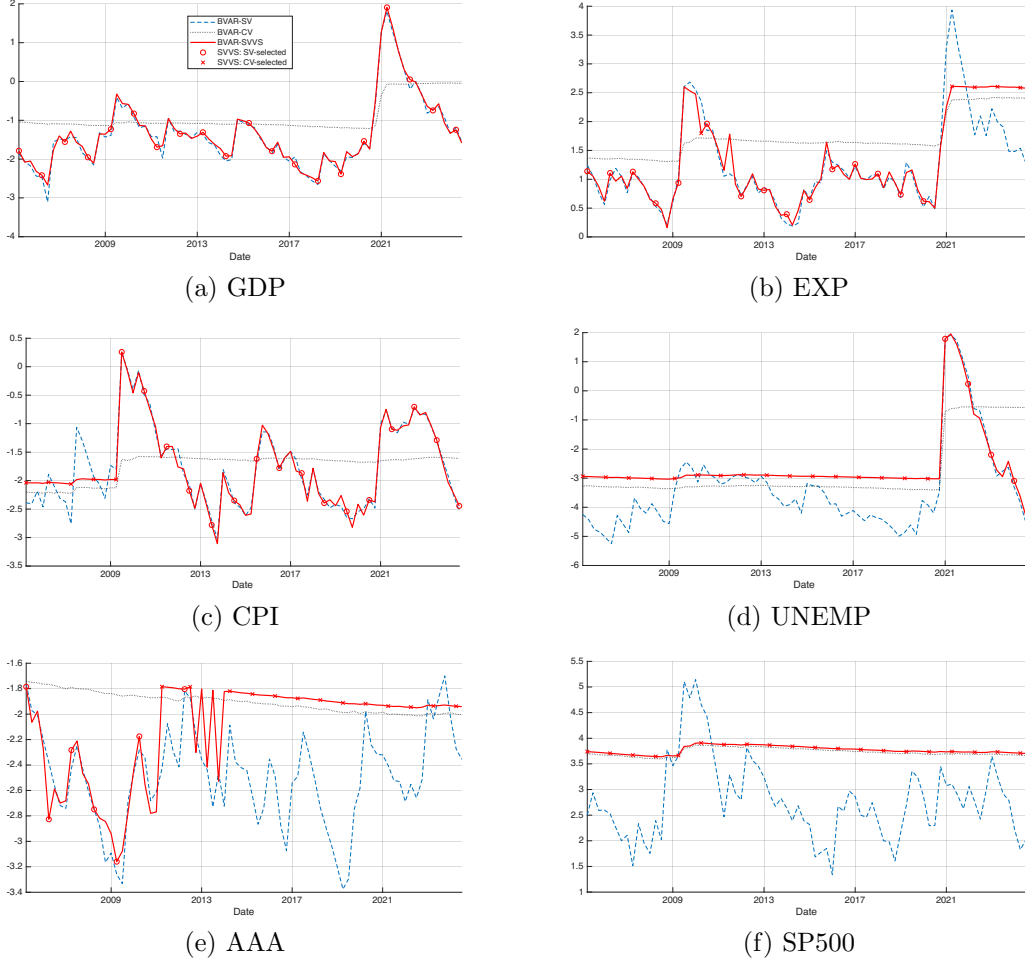
The table reveals a systematic horizon-parsimony tradeoff. Larger values of ν^{CV} shift more variables to the SV specification, producing more diffuse predictive densities that accommodate abrupt volatility episodes, which is an advantage at short horizons, while accumulating redundant second-moment variation at medium-to-long horizons. Smaller values do the reverse.

We set $\nu^{CV} = 15$ as the baseline. As Table 1 shows, this value yields the best joint ALPS at $h = 4$ across the entire grid and remains competitive at $h = 1$, $h = 8$, and $h = 16$, making it the configuration that most evenly distributes the horizon-parsimony trade-off. Researchers who prioritize short-horizon density accuracy may prefer larger values of ν^{CV} , which assign stochastic volatility to more variables; those targeting medium-to-long horizons may find smaller values more appropriate. The grid in Table 1 provides a direct map from ν^{CV} to the resulting classification and forecast profile, enabling practitioners to calibrate to their objectives.

Figure 1 plots forecast-implied log-variance paths for selected variables at horizon $h = 4$, comparing BVAR-SVVS with the BVAR-SV and BVAR-CV benchmarks. The paths are computed recursively across forecast origins and therefore summarize how predictive second moments are updated as the estimation sample expands. They should not be interpreted as within-window time variation in the specification indicator. Rather, BVAR-SVVS re-evaluates the relevance of the SV component at each forecast origin.

The figure shows broad heterogeneity across variables. GDP remains close to the BVAR-SV benchmark over much of the sample, suggesting persistent time variation in predictive uncertainty. Unemployment shifts sharply toward the SV benchmark around the COVID-19 period, while CPI moves from a relatively CV-like path in the earlier sample toward a more SV-like path after the Global Financial Crisis. Exports provide a useful contrast. After the COVID-19 period, their SVVS path moves closer

Figure 1.: Forecast-Implied Log-Variance Paths Across Models ($h = 4$)



to the CV benchmark, indicating that large shocks do not necessarily imply persistent stochastic volatility. What matters for classification is whether the shock generates forecast-relevant second-moment dynamics.

The financial variables display similarly mixed patterns. The Aaa corporate yield alternates between the SV and CV benchmarks across episodes, whereas the S&P 500 often remains closer to the CV benchmark. High observed volatility in financial markets does not automatically translate into persistent stochastic volatility relevant for quarterly density forecasting. Overall, the figure supports the central premise of BVAR-SVVS. The relevance of stochastic volatility is variable-specific and episode-dependent, so imposing SV uniformly can be unnecessarily restrictive.

5.3. Point Forecasting

Because point forecasts depend primarily on the conditional mean, RMSFE differences across volatility specifications are expected to be modest. Table 2 reports the relative RMSFE of BVAR-SVVS against the two benchmarks for the large system, with values below one indicating that BVAR-SVVS delivers lower RMSFE. Against BVAR-SV, most ratios are close to one across variables and horizons. The comparison against BVAR-CV is more favorable; BVAR-SVVS frequently produces ratios below one, particularly for interest-rate and credit-market variables (TR5, TR10, AAA, BAA), with several differences significant under the Diebold–Mariano test. Some real activity and price variables also improve over BVAR-CV, though less uniformly across horizons.

Table 2.: Relative RMSFE of BVAR-SVVS against BVAR-SV and BVAR-CV : Large Model

Variable	SVVS / SV				SVVS / CV			
	$h = 1$	$h = 4$	$h = 8$	$h = 16$	$h = 1$	$h = 4$	$h = 8$	$h = 16$
GDP	1.0032	1.0021	1.0041	0.9934*	0.7797	0.9875	0.9916	0.9935**
CONS	1.0004	1.0039	1.0030	1.0074	0.7570	0.9679	1.0017	0.9891**
INV	1.0889	1.0122	1.0039	1.0068	0.8152	0.9884	0.9394	1.0000
EXP	1.1633	1.0123	0.9859***	0.9866*	0.9640	0.9870	1.0138	1.0149
IMP	1.0623	0.9910	0.9940	0.9935	0.8869	0.9986	0.9866	0.9972
GOV	1.0044	1.0089	1.0186	1.0114	0.9351	0.9748	0.8849	0.9768
DISP	0.9959	0.9996	1.0018	0.9961	0.9923	0.9800	0.9868	0.9991
GDPP	1.0381	1.1026	1.1494	1.0813	0.9392	0.9671	1.0106	1.0080
PCE	1.0153	1.0335	1.0311	1.0290	0.9742	0.9867	0.9715***	0.9974
PCEX	0.9982	1.0376	1.0485	1.0170	0.9388	0.9567	0.9770	0.9848*
CPI	1.0083	1.0134	1.0081	1.0212	0.9567	0.9440	0.9524**	0.9943
POP	0.9975	0.9972	0.9993	1.0067	0.7914	0.8813	1.0169	1.0107
UEMP	0.9979	1.0175	0.9960	0.9894	0.7563	0.7289	0.8980	1.0272
WAGE	1.0003	0.9961	1.0031	0.9995	0.9578	0.9974	1.0171	1.0111
HOUS	1.0492	1.0375	1.0169	1.0156	0.9991	0.9368	0.9932	1.0204
IPI	1.0016	0.9990	0.9957	0.9878*	0.8773	0.9891	0.9727	0.9982
CAP	0.9961	0.9970	1.0123	0.9878***	0.8599	0.9814	0.9541	0.9895
SENT	0.9869	0.9338	0.9136	0.8362	0.9849	0.9647	1.0119	1.0067
TR2	0.9887	0.9835	0.9398**	0.8818**	0.8229	0.8094	0.9493	0.9655
TR5	1.0068	0.9880	0.9475**	0.9207*	0.8700*	0.7824*	0.8553	0.8825**
TR10	0.9885**	0.9743*	0.9562**	0.9535	0.9002**	0.7594**	0.7987	0.8343**
AAA	1.0016	1.0470	1.0385	0.9823	0.9340**	0.8245**	0.7859*	0.8208**
BAA	1.0014	0.9452	0.9716	0.9441	0.9529***	0.8354**	0.7565**	0.7727**
SP500	1.0106	1.0057	1.0222	1.0111	0.9632*	0.9858	0.9735	1.0017
WTI	1.0357	0.9972	0.9971	0.9869**	0.9757	1.0083	1.0024	1.0097

Note: This table reports the relative RMSFE of BVAR-SVVS against BVAR-SV and BVAR-CV across forecast horizons $h = \{1, 4, 8, 16\}$. Values below one indicate superior point forecast accuracy of BVAR-SVVS relative to the benchmark model. Superscripts *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively, based on the Diebold–Mariano test.

Results for the small and medium systems, reported in Appendix D.1, exhibit qualitatively similar patterns. Overall, the point-forecast gains from BVAR-SVVS are moderate and heterogeneous, broadly tracking BVAR-SV and improving on BVAR-CV mostly for interest-rate and credit-market variables. Volatility selection thus delivers its principal gains in density forecasting.

5.4. Density Forecasting

Table 3 reports joint ALPS differentials for BVAR-SVVS relative to the BVAR-SV and BVAR-CV benchmarks. In contrast to the RMSFE results, density forecasting provides clearer and more systematic evidence of gains from volatility selection, particularly in larger systems and at medium horizons.

Table 3.: Joint ALPS differentials over the out-of-sample period

(a) BVAR-SVVS relative to BVAR-SV					(b) BVAR-SVVS relative to BVAR-CV				
Dataset	$h = 1$	$h = 4$	$h = 8$	$h = 16$	Dataset	$h = 1$	$h = 4$	$h = 8$	$h = 16$
Large	-11.74	9.49	36.65	17.10	Large	24.43	11.52	-1.31	-10.68
Medium	-8.77	-6.26	2.79	2.96	Medium	7.73	6.45	1.88	1.10
Small	-0.25	-0.50	-0.29	-0.17	Small	2.01	1.49	1.30	1.33

Note: Each entry reports the difference in joint average log predictive scores (joint ALPS) over the out-of-sample evaluation period. Positive values indicate that BVAR-SVVS delivers a higher predictive likelihood than the corresponding benchmark and therefore provides superior density forecast performance.

Against BVAR-SV, BVAR-SVVS underperforms in the small system across all horizons, in the medium system at short horizons ($h = 1, 4$), and in the large system at $h = 1$. However, it outperforms BVAR-SV at medium-to-long horizons in the medium and large systems. These results support the parsimony argument that variable-specific selection avoids the unnecessary flexibility of imposing stochastic volatility on every variable.

The short-horizon loss in the large system stems from the role of extreme volatility

episodes in the evaluation sample. By assigning stochastic volatility to every variable, BVAR-SV generates more diffuse predictive densities during abrupt shocks such as the COVID-19 pandemic, which is advantageous at short horizons but less efficient on average at medium and longer horizons.

Comparison with BVAR-CV confirms that time-varying volatility matters—but only for a subset of the system. BVAR-SVVS improves on BVAR-CV in most cases, including all horizons in the small and medium systems and the short horizons in the large system, with negative differentials only at $h = 8$ and $h = 16$ in the large system. Marginal and cumulative score decompositions are reported in Appendices D5–D12.

Taken together, the density forecasting results show that the gains from BVAR-SVVS are selective rather than uniform. They are strongest in the large system at medium horizons, where variable-specific selection strikes a better balance between flexibility and parsimony than either benchmark alone.

5.5. Decomposing Density Forecasting

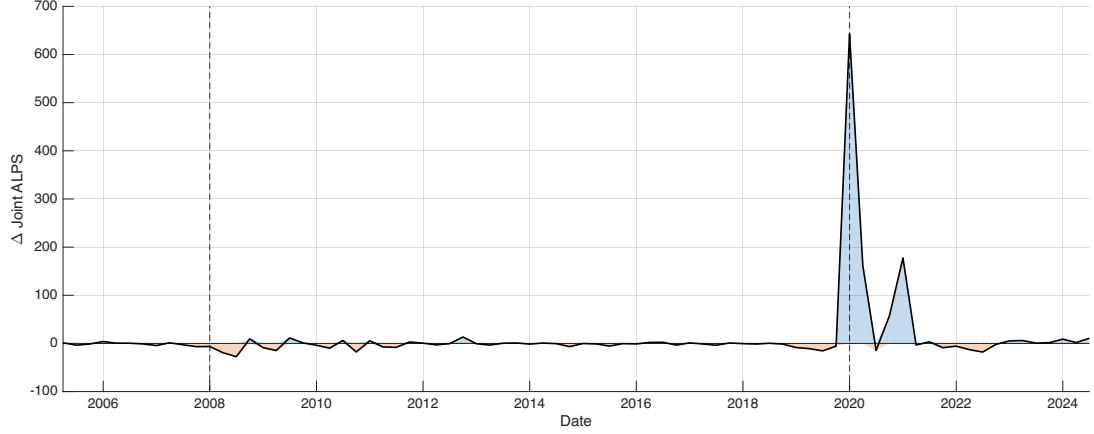
To trace the source of density-forecasting gains, we examine time-varying joint ALPS differentials. For benchmark m ,

$$\Delta \text{ALPS}_{t,h}^m = \log p^{\text{BVAR-SVVS}}(y_{t+h} \mid \mathcal{I}_t) - \log p^m(y_{t+h} \mid \mathcal{I}_t), \quad (5.5)$$

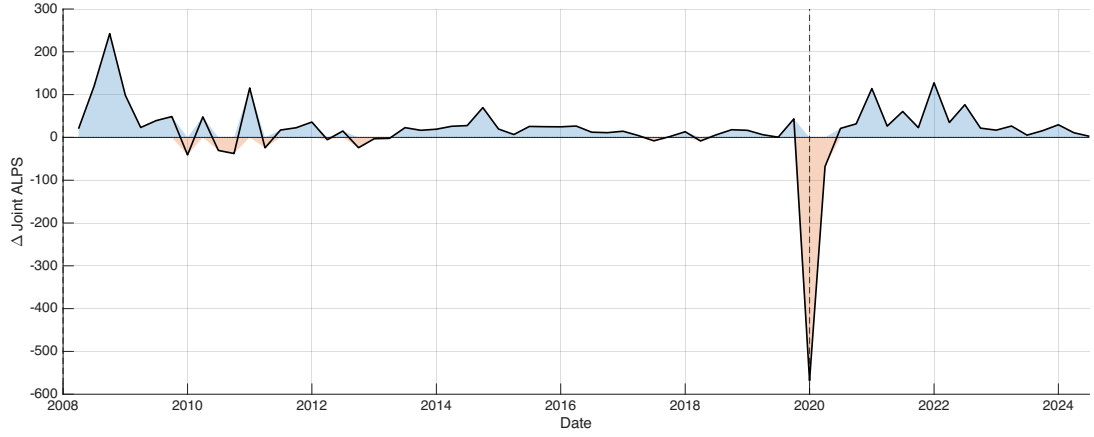
with positive values indicating that BVAR-SVVS assigns higher joint predictive density to the realized outcome at forecast origin t . Figure 2 plots these differentials for two representative comparisons.

Panel 2(a), comparing BVAR-SVVS with BVAR-CV at $h = 4$, shows the largest positive differentials around the COVID-19 period, when the homoskedastic benchmark cannot adjust predictive uncertainty as it shifts abruptly. Panel 2(b), comparing

Figure 2.: Time-varying ALPS differentials



(a) BVAR-SVVS vs. BVAR-CV ($h = 4$)

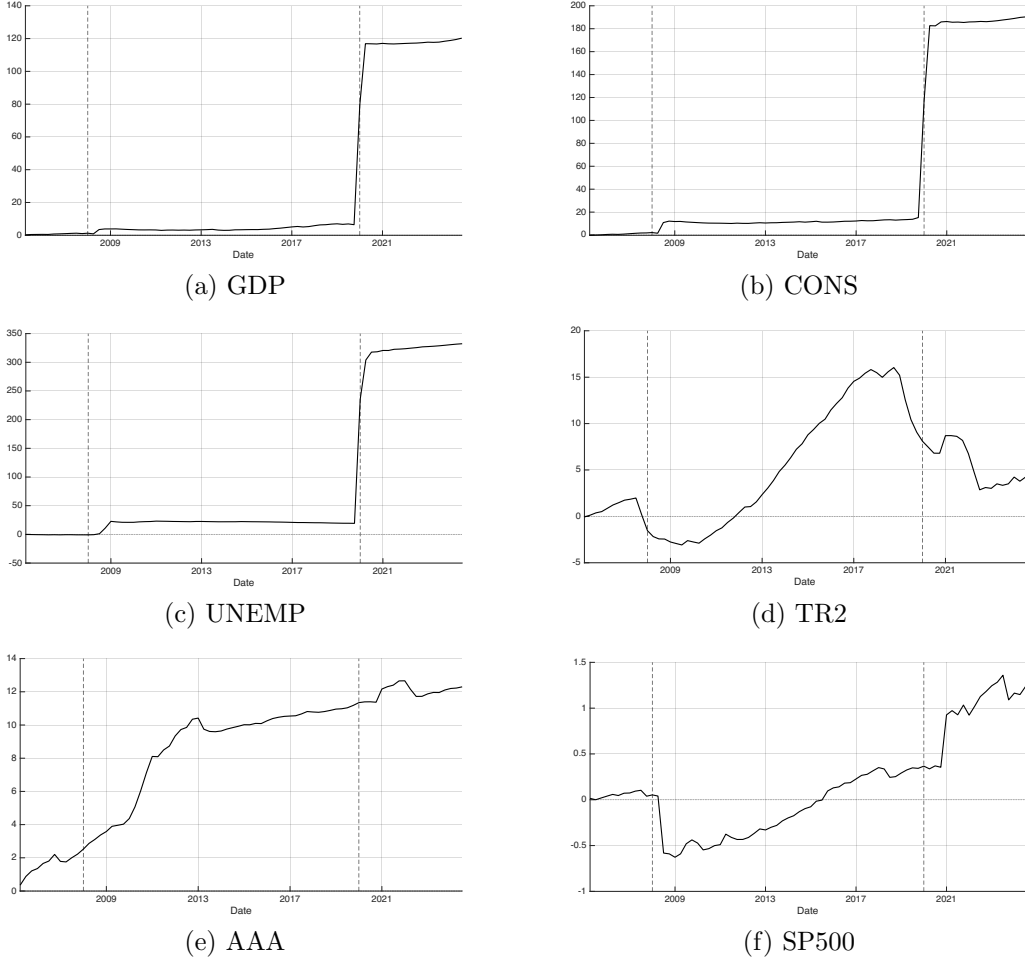


(b) BVAR-SVVS vs. BVAR-SV ($h = 16$)

with BVAR-SV at $h = 16$, reverses this pattern—BVAR-SVVS performs better outside crisis periods but sharply worse during the COVID peak. These Figures reveal two complementary channels of gain. BVAR-SVVS captures the volatility episodes that BVAR-CV misses and avoids the unnecessary volatility dynamics that BVAR-SV imposes during stable periods.

Figure 3 reports cumulative marginal LPS differentials of BVAR-SVVS relative to BVAR-CV at $h = 4$ for variables spanning real activity, labor markets, interest rates, credit markets, and financial markets. Positive values indicate higher cumulative marginal predictive scores under BVAR-SVVS. The gains are especially pronounced

Figure 3.: Cumulative Marginal LPS: BVAR-SVVS vs. BVAR-CV ($h = 4$)

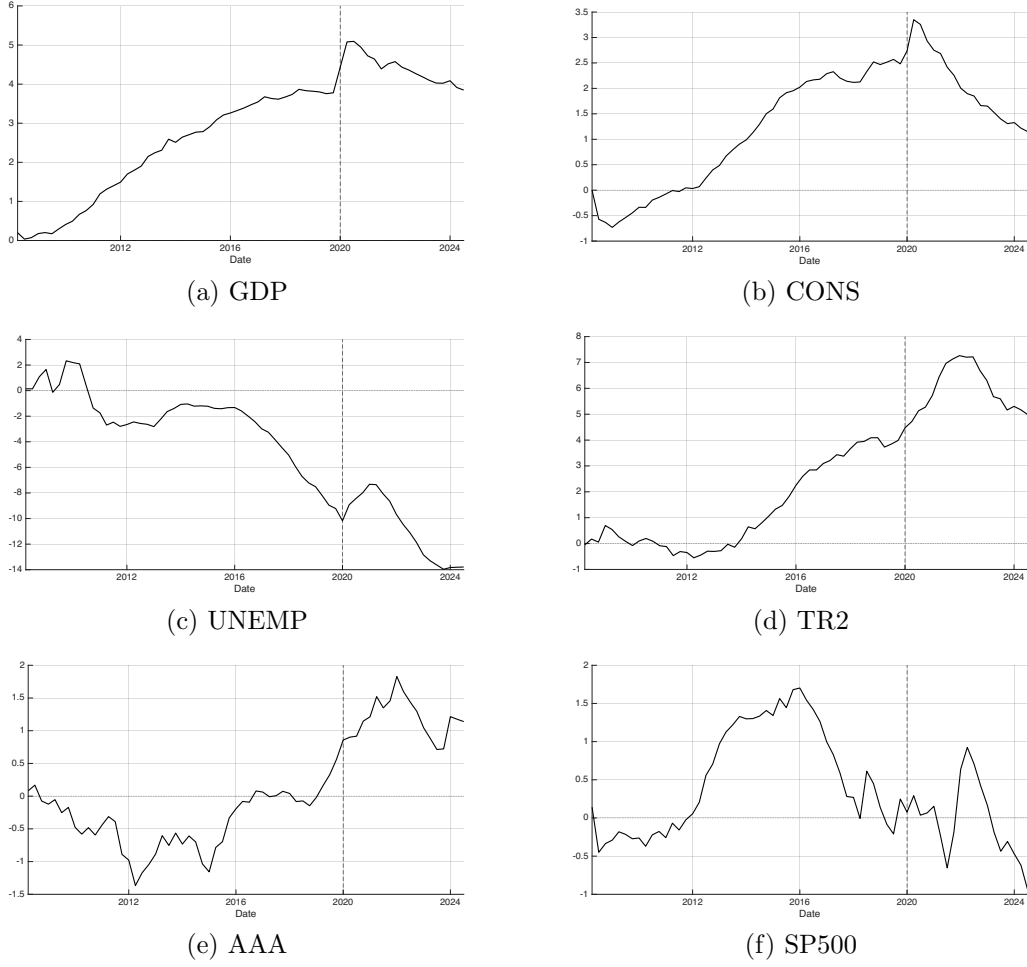


for real activity and labor-market variables around the COVID-19 period—for GDP, consumption, and unemployment, the cumulative differentials rise sharply during this episode, consistent with the inability of a constant-volatility specification to track abrupt macroeconomic disruptions.

Financial and interest-rate variables show different dynamics. The two-year Treasury rate, the Aaa corporate yield, and the S&P 500 display cumulative improvements that accumulate gradually rather than concentrating in a single episode. The improvement over BVAR-CV is therefore both variable-specific and time-varying.

Figure 4 reports cumulative marginal LPS differentials of BVAR-SVVS relative to BVAR-SV at $h = 16$, isolating the value of selecting stochastic volatility variable by

Figure 4.: Cumulative Marginal LPS: BVAR-SVVS vs. BVAR-SV ($h = 16$)



variable rather than imposing it uniformly. For GDP and consumption, the cumulative differentials are positive over most of the evaluation period and rise sharply around the COVID-19 episode. The two-year Treasury rate also shows sustained gains from the mid-2010s onward, while the Aaa corporate yield turns positive around 2019 and continues to improve through the post-COVID period. These patterns indicate that BVAR-SVVS can outperform the BVAR-SV benchmark at longer horizons by avoiding excessive second-moment variation when stochastic volatility is not uniformly needed.

The gains are not uniform, however. The unemployment rate exhibits a persistently negative cumulative differential, with BVAR-SV performing better for this variable over much of the sample. The S&P 500 displays a more mixed pattern. Its cumulative

differential rises strongly through the mid-2010s and remains positive for a substantial part of the evaluation period, although the advantage weakens toward the end of the sample. This heterogeneity is consistent with the premise that the relevance of stochastic volatility differs across variables and over time.

Importantly, these marginal gains need not map one for one into a variable’s own SV/CV classification. Even when a variable receives the same volatility specification under BVAR-SVVS and a benchmark, its predictive density can change because the volatility specifications of other variables alter the system-wide posterior estimates. Thus, the decomposition captures both direct gains from variable-level volatility selection and indirect gains from estimating a more parsimonious joint system.

6. Robustness to Contemporaneous Dependence

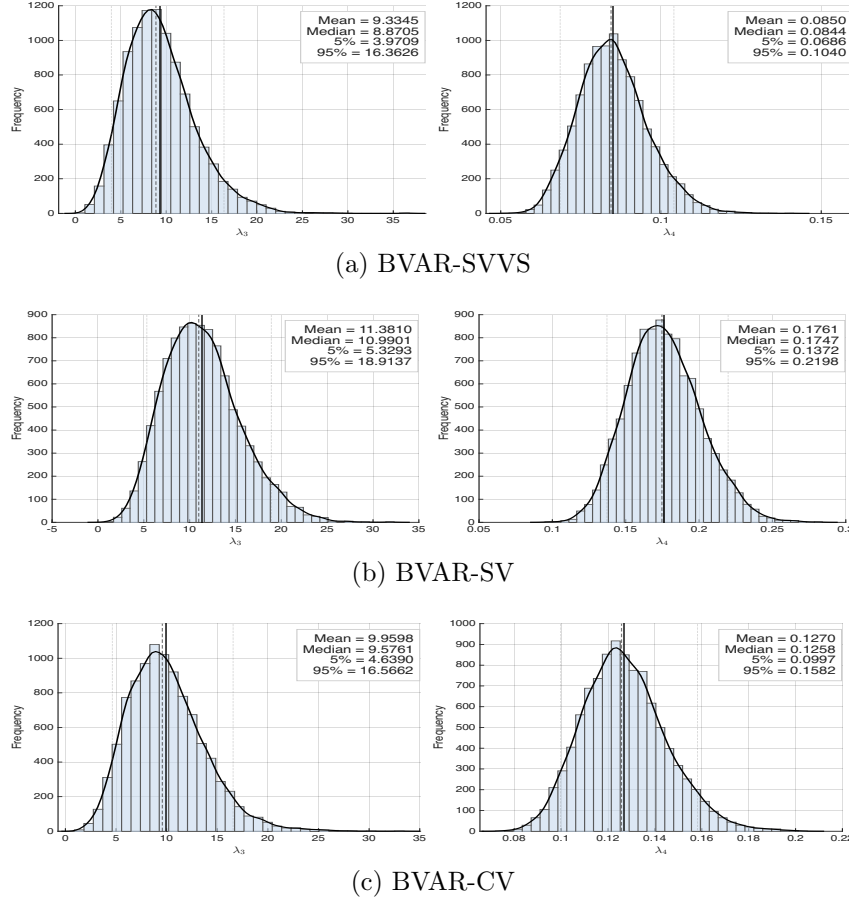
The baseline specification imposes strong shrinkage on the contemporaneous impact matrix to focus the comparison on the volatility specification. We now examine whether the main findings survive a more flexible treatment of contemporaneous dependence by replacing the fixed prior scales for the intercept and contemporaneous coefficients with hierarchical priors,

$$\lambda_3 \sim \mathcal{G}(e_{31}, e_{32}) \text{ and } \lambda_4 \sim \mathcal{G}(e_{41}, e_{42}), \quad (6.1)$$

where $e_{31}, e_{32}, e_{41}, e_{42}$ are shape and rate hyperparameters, all set to one. This treatment is applied symmetrically to BVAR-SVVS and the benchmark models, allowing the degree of shrinkage to adjust endogenously across specifications.⁵ The sampling procedure is detailed in Appendix C.8.

⁵All other hyperparameters retain their baseline values. The grid search over ν^{CV} was repeated under this specification with essentially unchanged classifications (Table B1 in Appendix B).

Figure 5.: Posterior Distributions of λ_3 and λ_4 in the Large Model



Note: The figure reports posterior frequency distributions of λ_4 (left column) and λ_3 (right column) for the Large model under BVAR-SVVS, BVAR-SV, and BVAR-CV. In our specification, λ_4 controls the degree of shrinkage on contemporaneous impact coefficients, whereas λ_3 governs the prior scale for intercept terms.

Figure 5 reports the posterior distributions of λ_3 and λ_4 for the large system.⁶ The posterior of λ_3 , which scales the intercept prior, is broadly similar across BVAR-SVVS, BVAR-SV, and BVAR-CV. This suggests that intercept shrinkage is largely insensitive to the volatility specification. The posterior of λ_4 , which scales the contemporaneous-coefficient prior, differs visibly. Its mean is largest under BVAR-SV, smaller under BVAR-CV, and smallest under BVAR-SVVS. The posterior therefore imposes the tightest contemporaneous-dependence prior on BVAR-SVVS and the loosest on BVAR-SV.

⁶Appendix D.6 reports analogous results for the small and medium systems.

This pattern reflects how the models allocate second-moment variation. BVAR-SV imposes stochastic volatility uniformly and tends to rely more on contemporaneous dependence to fit joint predictive variation. BVAR-SVVS, by contrast, uses variable-specific selection for a more parsimonious volatility structure, and the contemporaneous block is correspondingly shrunk more tightly. BVAR-CV lies between, avoiding latent volatility variation but lacking direct flexibility for time-varying uncertainty.

Table 4.: Average Joint Predictive Likelihoods over the Out-of-Sample Period with Estimated Covariance Structures

(a) BVAR-SVVS relative to BVAR-SV					(b) BVAR-SVVS relative to BVAR-CV				
Dataset	$h = 1$	$h = 4$	$h = 8$	$h = 16$	Dataset	$h = 1$	$h = 4$	$h = 8$	$h = 16$
Large	3.87	27.15	48.18	33.94	Large	11.15	10.78	-2.82	-3.17
Medium	-3.35	0.12	2.30	5.36	Medium	6.25	6.53	-1.61	2.60
Small	-0.15	-0.21	-0.29	-0.06	Small	2.04	1.85	1.45	1.70

Note: Each entry reports the joint predictive likelihood differential averaged over the out-of-sample evaluation period. Positive values indicate superior density forecast performance of BVAR-SVVS relative to the corresponding benchmark. The reported values are based on the estimated covariance structures.

Table 4 reports joint ALPS differentials under the hierarchical priors on intercept and contemporaneous-coefficient shrinkage. Allowing flexible contemporaneous dependence strengthens the performance of BVAR-SVVS mainly against BVAR-SV. The improvement is most evident in the large system, where BVAR-SVVS now outperforms BVAR-SV at all horizons. Once contemporaneous dependence is estimated, imposing stochastic volatility uniformly becomes less attractive in large systems.

The improvement is not uniform across system sizes. BVAR-SV still performs slightly better at $h = 1$ in the medium system and in several small-system cases, consistent with the view that the benefits of variable-specific volatility selection are more pronounced when the model dimension is large. In smaller systems, imposing stochastic volatility on every variable carries a smaller cost, and the gains from volatility parsimony are correspondingly weaker.

Against BVAR-CV, the results are more mixed. BVAR-SVVS continues to outperform in several cases, but the gains are smaller than in the baseline and turn negative in the large system at $h = 8$ and $h = 16$. Estimating contemporaneous dependence thus strengthens BVAR-SVVS against BVAR-SV but does not uniformly improve its performance against BVAR-CV.

Overall, the robustness exercise supports the main conclusion that variable-specific volatility selection is most valuable in larger systems. At the same time, the results should be interpreted as supplementary evidence, since lower-triangular covariance specifications remain sensitive to variable ordering and do not fully resolve identification issues in the contemporaneous block.

7. Conclusion

The central finding is that volatility specification is not a system-level property; it is variable-specific. Real activity series support stochastic volatility, while financial and price-related series often do not. Treating the variance equation as an object of model selection, rather than an assumption imposed uniformly, recovers this heterogeneity and converts it into predictive gains.

The density-forecasting improvements are concentrated where parsimony has the most bite: large systems at medium-to-long horizons, where redundant volatility dynamics compound across equations and erode predictive calibration. Point forecasts improve more selectively, a natural consequence of volatility selection operating on the second moment rather than the conditional mean. Relaxing strong shrinkage on the contemporaneous impact matrix strengthens the case for BVAR-SVVS against BVAR-SV in large systems, though results against BVAR-CV become more mixed. The main qualitative conclusions are robust to this extension.

Three directions stand out for future work. First, the Cholesky-based factorization

creates sensitivity to variable ordering. An order-invariant covariance specification would remove this arbitrariness. Second, ν^{CV} is currently calibrated by grid search. Estimating it within the Gibbs sampler would complete the hierarchical design and eliminate the last manually tuned element. Third, the logic of variable-specific selection extends naturally beyond the variance equation. Lag length, prior tightness, and mean-equation structure could in principle all be selected variable by variable within the same framework. The present paper addresses the second moment. The broader program of variable-specific Bayesian regularization in large VARs remains largely open.

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Online Appendices:

Stochastic Volatility Selection in Bayesian Vector Autoregressions

Jingyu Song and Kyu Ho Kang

Appendix A. Data

Table A1.: Data Description

Variable	Mnemonic	Transf.	Small	Medium
Gross Domestic Product	GDP	QoQ	✓	✓
Personal Consumption Expenditures	CONS	QoQ		✓
Private Domestic Investment	INV	QoQ		✓
Exports	EXP	QoQ		✓
Imports	IMP	QoQ		
Government Expenditure	GOV	QoQ		
Disposable Personal Income	DISP	QoQ		
GDP Price Index	GDPP	QoQ		
PCE Price Index	PCE	QoQ		
Core PCE Price Index	PCEX	QoQ	✓	✓
Consumer Price Index	CPI	QoQ		
Core CPI	CPIX	QoQ		
All Employees: Nonfarm	POP	QoQ		
Unemployment Rate	UEMP	Level		✓
Real Hourly Compensation	WAGE	QoQ		
Housing Starts	HOUS	QoQ		✓
Industrial Production Index	IPI	QoQ		
Capacity Utilization	CAP	Level		
Consumer Sentiment	SENT	Level		✓
2-Year Treasury Yield	TR2	Level	✓	✓
5-Year Treasury Yield	TR5	Level		
10-Year Treasury Yield	TR10	Level		✓
Moody's Aaa Corporate Bond	AAA	Level		
Moody's Baa Corporate Bond	BAA	Level		
S&P 500 Composite Index	SP500	QoQ		✓
WTI Crude Oil Price	WTI	QoQ		

Note: “QoQ” denotes the quarterly log difference, $100 \times \Delta \log y_t$, while “Level” indicates variables used without transformation. Real activity variables are measured in billions of chained 2017 U.S. dollars; price indexes are index series; interest rates and corporate bond yields are measured in percent. A checkmark (✓) indicates inclusion in the corresponding model specification. The large model includes all variables by construction.

The empirical analysis uses a quarterly U.S. dataset of 25 macroeconomic and financial variables, following the data structure in Crump et al. (2025). As in that study, we exclude the federal funds rate to avoid complications associated with the zero lower

bound.⁷ The sample spans 1978Q1 to 2024Q3, and all variables are observed at the quarterly frequency. Most series enter the model as quarter-over-quarter log growth rates, while variables judged to be already stationary are included in levels.

To evaluate performance across different model sizes, we consider three systems of increasing dimension: a small system with 3 variables, a medium system with 10 variables, and a large system with 25 variables. The medium-sized system is based on Wright (2013), with minor adjustments to align it with the present forecasting design. Across all three cases, the variable sets are chosen to preserve broad macroeconomic coverage while including series with different volatility characteristics. This design is useful for assessing whether the proposed framework can distinguish between stochastic-volatility and constant-volatility processes across variables.

Table A1 summarizes the variable definitions and data transformations.

Appendix B. Grid Search under Estimated Contemporaneous Covariance

Table B1.: Classification of Constant and Stochastic Variables by ν_{CV} under Estimated Contemporaneous Covariance

S^{CV}	5.00	10.00	15.00	20.00	25.00	30.00	35.00	40.00	50.00	60.00	70.00	80.00	90.00	100.00
GDP														
CONS														
INV														
EXP														
IMP														
GOV														
DISP														
GDPP														
PCE														
PCEX														
CPI														
POP														
UEMP														
WAGE														
HOUS														
IPI														
CAP														
SENT														
TR2														
TR5														
TR10														
AAA														
BAA														
SP500														
WTI														
# CV	23.00	17.00	13.00	12.00	10.00	9.00	8.00	6.00	5.00	3.00	2.00	2.00	1.00	1.00
Joint ALPS ($h = 1$)	-49.82	-49.19	-48.03	-48.96	-51.34	-52.81	-53.30	-53.95	-60.62	-66.81	-69.84	-66.40	-61.16	-55.57
Avg. RMSFE ($h = 1$)	3.12	3.09	3.09	3.11	3.10	3.07	3.06	3.07	3.08	3.08	3.07	3.08	3.10	3.09
Joint ALPS ($h = 4$)	-63.54	-74.95	-80.08	-85.21	-92.72	-93.05	-97.44	-98.43	-119.73	-116.02	-111.81	-113.76	-133.38	-129.60
Avg. RMSFE ($h = 4$)	1853.37	3.81	3.38	3.39	3.38	3.38	3.39	3.41	3.41	3.43	3.44	3.41	3.45	3.42
Joint ALPS ($h = 8$)	-76.70	-80.43	-89.59	-91.37	-107.85	-109.39	-108.60	-115.00	-123.92	-144.27	-146.27	-152.54	-159.99	-138.98
Avg. RMSFE ($h = 8$)	2095.95	3.95	3.68	3.67	3.69	3.69	3.70	3.70	3.77	3.74	3.73	3.72	3.74	3.74
Joint ALPS ($h = 16$)	-80.64	-87.50	-91.72	-97.02	-111.41	-105.89	-105.39	-118.61	-138.82	-132.10	-149.76	-146.00	-139.59	-156.18
Avg. RMSFE ($h = 16$)	159.74	4.29	3.93	3.94	3.93	3.93	3.96	3.97	4.04	4.10	4.04	4.05	4.03	4.03

Note: Shaded cells denote CV specifications, while blank cells correspond to SV specifications. The classifications are based on full-sample estimation and summarize how the mixture composition varies with S^{CV} . In the rolling forecasting exercise, the selected classification may change over time as the information set expands; see Figure 1. The first row reports the number of variables classified as CV. Joint ALPS and average RMSFE are reported for forecast horizons $h = \{1, 4, 8, 16\}$.

⁷We also exclude the S&P GSCI Non-energy Index, S&P 500 Realized Volatility, and disaggregated investment components. Although economically relevant, these series add limited incremental information for quarterly macroeconomic forecasting once aggregate investment, commodity prices, and financial yields are already included, while increasing sector-specific detail and model dimensionality.

Appendix C. Derivation of the Conditional Posterior

Appendix C.1. Sampling ϑ_i

The posterior distribution of the coefficient vector ϑ_i in equation i is obtained by combining the Gaussian likelihood with the Minnesota prior. Because both components are Gaussian, the conditional posterior is also Gaussian:

$$\vartheta_i \mid y, h, \lambda \sim \mathcal{N}(\hat{\vartheta}_i, K_{\vartheta_i}^{-1}),$$

where the posterior precision matrix and posterior mean are

$$\begin{aligned} K_{\vartheta_i} &= V_i^{-1} + X_i' H_i^{-1} X_i, \\ \hat{\vartheta}_i &= K_{\vartheta_i}^{-1} (V_i^{-1} m_i + X_i' H_i^{-1} y_i). \end{aligned}$$

In this specification, ϑ_i is a $k_i \times 1$ vector of coefficients with $k_i = np + i$. It comprises $np + 1$ parameters for the intercept and lagged regressors, and $i - 1$ parameters for the contemporaneous regressors. The dependent variable is collected in the vector $y_i = (y_{i,1}, \dots, y_{i,T})'$, and the $T \times k_i$ regressor matrix X_i is formed by stacking the row vectors $x_{i,t}$ for $t = 1, \dots, T$. The time-varying error variances are summarized in the diagonal matrix $H_i = \text{diag}(e^{h_{i,1}}, \dots, e^{h_{i,T}})$. The prior mean vector m_i and the diagonal prior covariance matrix V_i are defined as

$$V_i = \text{diag}\left(\lambda_3, v_{1,i,1}, \dots, v_{1,i,n}, \dots, v_{p,i,1}, \dots, v_{p,i,n}, \underbrace{\lambda_4, \dots, \lambda_4}_{i-1 \text{ times}}\right)$$

Direct sampling requires repeated inversion of the $k_i \times k_i$ precision matrix K_{ϑ_i} at each Gibbs iteration, which becomes computationally intensive in high-dimensional settings. We therefore follow the precision-based sampling approach of Chan (2020).⁸ Together, these steps leverage the sparse structure of the precision matrix and reduce the computational burden as n grows.

Appendix C.2. Sampling λ

The shrinkage hyperparameters λ are updated by combining the Gaussian prior kernel implied by the Minnesota prior with the Gamma hyperpriors in (2.9). Let S_1 denote the set of lag coefficients whose prior variances depend on λ_1 . For $(l, i, j) \in S_1$, write $v_{l,i,j} = \lambda_1 \tilde{v}_{l,i,j}$, where $\tilde{v}_{l,i,j}$ collects the remaining scale terms that do not depend on λ_1 . Then the conditional posterior kernel for λ_1 is

$$\begin{aligned} \pi(\lambda_1 \mid \vartheta) &\propto \prod_{(l,i,j) \in S_1} (\lambda_1 \tilde{v}_{l,i,j})^{-1/2} \exp\left(-\frac{1}{2} \frac{(\vartheta_{l,i,j} - m_{l,i,j})^2}{\lambda_1 \tilde{v}_{l,i,j}}\right) \times \lambda_1^{e_{11}-1} \exp(-e_{12}\lambda_1) \\ &\propto \lambda_1^{-\frac{|S_1|}{2} + e_{11} - 1} \exp\left[-\frac{1}{2} \left(2e_{12}\lambda_1 + \frac{1}{\lambda_1} \sum_{(l,i,j) \in S_1} \frac{(\vartheta_{l,i,j} - m_{l,i,j})^2}{\tilde{v}_{l,i,j}}\right)\right]. \end{aligned}$$

This kernel is recognizable as a Generalized Inverse Gaussian (GIG) distribution.

Hence the full conditional distribution of λ_1 is given by

$$\lambda_1 \mid \vartheta \sim \text{GIG}\left(-\frac{|S_1|}{2} + e_{11}, 2e_{12}, \sum_{(l,i,j) \in S_1} \frac{(\vartheta_{l,i,j} - m_{l,i,j})^2}{\tilde{v}_{l,i,j}}\right).$$

Similarly, let S_2 denote the set of cross-variable lag coefficients, and write $v_{l,i,j} =$

⁸Specifically, let C denote the lower-triangular Cholesky factor of K_{ϑ_i} such that $K_{\vartheta_i} = CC'$. We first solve $Cw = (V_i^{-1}m_i + X_i'H_i^{-1}y_i)$ for w via forward substitution, and then obtain the posterior mean $\hat{\vartheta}_i$ by solving $C'\hat{\vartheta}_i = w$ via backward substitution. To generate a draw for ϑ_i , we simulate $u \sim \mathcal{N}(0, I_{k_i})$ and compute $\vartheta_i = \hat{\vartheta}_i + (C')^{-1}u$, which avoids the explicit computation of the inverse matrix $K_{\vartheta_i}^{-1}$.

$\lambda_1 \lambda_2 \tilde{v}_{l,i,j}$ for $(l, i, j) \in S_2$. Conditional on λ_1 , the posterior kernel for λ_2 is

$$\begin{aligned} \pi(\lambda_2 \mid \vartheta, \lambda_1) &\propto \prod_{(l,i,j) \in S_2} (\lambda_1 \lambda_2 \tilde{v}_{l,i,j})^{-1/2} \exp\left(-\frac{1}{2} \frac{(\vartheta_{l,i,j} - m_{l,i,j})^2}{\lambda_1 \lambda_2 \tilde{v}_{l,i,j}}\right) \times \lambda_2^{e_{21}-1} \exp(-e_{22} \lambda_2) \\ &\propto \lambda_2^{-\frac{|S_2|}{2} + e_{21} - 1} \exp\left[-\frac{1}{2} \left(2e_{22} \lambda_2 + \frac{1}{\lambda_2} \sum_{(l,i,j) \in S_2} \frac{(\vartheta_{l,i,j} - m_{l,i,j})^2}{\lambda_1 \tilde{v}_{l,i,j}}\right)\right], \end{aligned}$$

so that we have

$$\lambda_2 \mid \vartheta, \lambda_1 \sim \text{GIG}\left(-\frac{|S_2|}{2} + e_{21}, 2e_{22}, \sum_{(l,i,j) \in S_2} \frac{(\vartheta_{l,i,j} - m_{l,i,j})^2}{\lambda_1 \tilde{v}_{l,i,j}}\right).$$

Regarding the contemporaneous impact coefficients, we impose strong prior shrinkage by fixing λ_4 close to zero in our baseline specification. This restriction isolates the role of volatility selection from contemporaneous interactions, which can otherwise complicate the interpretation of predictive densities in large systems (Chan et al., 2024). This approach is also prudent under the lower-triangular identification scheme, where estimation errors in earlier equations can propagate through the recursive structure and distort subsequent volatility estimates in high-dimensional models. In a subsequent extension, we relax this restriction and allow λ_4 to be determined hierarchically, enabling us to assess the robustness of our results when richer contemporaneous dependence is incorporated. We also set $\lambda_3 = 100$ to keep the intercept prior relatively diffuse.

Appendix C.3. Sampling α_i and ϕ_i

Conditional on the specification indicator $\gamma_i \in \{0, 1\}$, the log-volatility equation in (2.3) can be written as a Gaussian regression in $(\alpha_i, \phi_i)'$. Let $h_i^+ = (h_{i,2}, \dots, h_{i,T})'$ and $H_i = [\mathbf{1}_{T-1}, (h_{i,1}, \dots, h_{i,T-1})']$, where $\mathbf{1}_{T-1}$ is a $(T-1)$ -vector of ones. Under a

specification-dependent Gaussian prior,

$$(\alpha_i, \phi_i)' \mid \gamma_i \sim \mathcal{N}(\mu_{\gamma_i}, P_{\gamma_i}),$$

where μ_{γ_i} and P_{γ_i} denote the prior mean vector and prior covariance matrix associated with specification γ_i . Combining the Gaussian likelihood with the prior yields the conditional posterior distribution:

$$(\alpha_i, \phi_i)' \mid h_i, \sigma_{i,h}^2, \gamma_i \sim \mathcal{N}(\hat{\mu}_i, \hat{P}_i),$$

where

$$\begin{aligned} \hat{P}_i &= \left(\frac{1}{\sigma_{i,h}^2} H_i' H_i + P_{\gamma_i}^{-1} \right)^{-1}, \\ \hat{\mu}_i &= \hat{P}_i \left(\frac{1}{\sigma_{i,h}^2} H_i' h_i + P_{\gamma_i}^{-1} \mu_{\gamma_i} \right). \end{aligned}$$

Appendix C.4. Sampling $\sigma_{i,h}^2$

Conditional on (α_i, ϕ_i) , the innovation variance $\sigma_{i,h}^2$ is sampled from a specification-dependent Inverse-Gamma prior:

$$\sigma_{i,h}^2 \mid \gamma_i \sim \mathcal{IG}\left(\frac{\nu_{\gamma_i}}{2}, \frac{S_{\gamma_i}}{2}\right),$$

where $\nu_{\gamma_i} = \gamma_i \nu^{SV} + (1 - \gamma_i) \nu^{CV}$ and $S_{\gamma_i} = \gamma_i S^{SV} + (1 - \gamma_i) S^{CV}$. Combining this prior with the Gaussian likelihood implied by (2.3) yields the conjugate full conditional

distribution:

$$\sigma_{i,h}^2 \mid h_i, \alpha_i, \phi_i, \gamma_i \sim \mathcal{IG}\left(\frac{\hat{\nu}_i}{2}, \frac{\hat{S}_i}{2}\right),$$

where

$$\begin{aligned}\hat{\nu}_i &= \nu_{\gamma_i} + (T - 1), \\ \hat{S}_i &= S_{\gamma_i} + \sum_{t=2}^T (h_{i,t} - \alpha_i - \phi_i h_{i,t-1})^2.\end{aligned}$$

Thus, the specification indicator γ_i determines the prior on $\sigma_{i,h}^2$. When $\gamma_i = 0$, the prior is tightly concentrated around a constant-volatility specification. When $\gamma_i = 1$, it shifts toward a more flexible stochastic-volatility specification.

Appendix C.5. Sampling h_i

The vector of log-volatilities for each equation, $h_i = (h_{i,1}, \dots, h_{i,T})'$, is sampled from its full conditional posterior distribution. Conditional on the mixture indicators $s_i = (s_{i,1}, \dots, s_{i,T})'$, the model becomes conditionally Gaussian and can be expressed in a linear state-space form:

$$y_{i,t}^* = \ln[(y_{i,t} - x_{i,t}\vartheta_i)^2] = h_{i,t} + \epsilon_{i,t}^*, \quad \epsilon_{i,t}^* \sim \mathcal{N}(m_{s_{i,t}}, v_{s_{i,t}}^2), \quad (\text{C1})$$

$$h_{i,t} = \alpha_i + \phi_i h_{i,t-1} + \eta_{i,t}, \quad \eta_{i,t} \sim \mathcal{N}(0, \sigma_{i,h}^2). \quad (\text{C2})$$

Given the Gaussian structure in (C1)–(C2), the entire vector h_i can be efficiently drawn jointly from its multivariate Normal conditional posterior using the algorithm of Carter and Kohn (1994).

In the forward pass, the Kalman filter recursively computes the filtered means and

variances $(h_{t|t}, P_{t|t})$ for $t = 1, \dots, T$. In the backward pass, h_t is sequentially drawn for $t = T - 1, \dots, 1$ from

$$h_{i,t} \mid h_{i,t+1}, y_{1:T}, s_i, \vartheta_i \sim \mathcal{N}(h_{t|t, h_{t+1}}, P_{t|t, h_{t+1}}), \quad (\text{C3})$$

where

$$h_{t|t, h_{t+1}} = h_{t|t} + P_{t|t} \phi_i (P_{t+1|t})^{-1} (h_{t+1} - h_{t+1|t}), \quad (\text{C4})$$

$$P_{t|t, h_{t+1}} = P_{t|t} - P_{t|t} \phi_i^2 (P_{t+1|t})^{-1} P_{t|t}. \quad (\text{C5})$$

The forward-filtered quantities $(h_{t|t}, P_{t|t})$ and $(h_{t+1|t}, P_{t+1|t})$ are obtained from standard Kalman recursions. Because the algorithm is now standard in the stochastic volatility literature, these details are omitted here for brevity (see Kim et al., 1998; Omori et al., 2007).

Appendix C.6. Sampling s_t (Mixture Component Indicators)

Following Kim et al. (1998) and Omori et al. (2007), the non-Gaussian log-squared error term is approximated by a finite mixture of K Normal distributions:

$$\sum_{i=1}^K \pi_i \mathcal{N}(m_i, v_i^2), \quad (\text{C6})$$

where π_i , m_i , and v_i^2 are fixed mixture weights, means, and variances.⁹

Given this representation, the indicator variable $s_t \in \{1, \dots, K\}$ specifies the Normal component associated with observation t . The conditional posterior probability

⁹We adopt the 10-component Normal mixture approximation of Omori et al. (2007), which achieves near-exact fidelity to the non-Gaussian log-squared error distribution.

that $S_t = i$ is given by

$$\Pr(S_t = i \mid y, h, \sigma_h^2) = \frac{\pi_i \mathcal{N}(y_t^* \mid h_t + m_i, v_i^2)}{\sum_{q=1}^K \pi_q \mathcal{N}(y_t^* \mid h_t + m_q, v_q^2)}. \quad (\text{C7})$$

At each iteration of the Gibbs sampler, these normalized probabilities are computed for all $i = 1, \dots, K$, and S_t is drawn from a categorical distribution with probabilities proportional to (C7). This step effectively linearizes the measurement equation, thereby allowing the volatility states h_i to be sampled using the Gaussian FFBS algorithm in the subsequent step.

Appendix C.7. Sampling p

The parameter p represents the prior inclusion probability for stochastic volatility, governing the overall tendency of the model to favor the stochastic-volatility (SV) specification over the constant-volatility (CV) specification. Given the current draws of the specification indicators $\gamma = (\gamma_1, \dots, \gamma_n)'$, the posterior distribution of p follows directly from the Beta–Bernoulli conjugacy.

The joint likelihood of γ given p is

$$p(\gamma \mid p) = \prod_{k=1}^n p^{\gamma_k} (1 - p)^{1 - \gamma_k}. \quad (\text{C8})$$

Assuming the Beta prior

$$p \sim \text{Beta}(a_p, b_p), \quad (\text{C9})$$

the resulting posterior distribution is also Beta, with updated shape parameters:

$$p \mid \gamma \sim \text{Beta}\left(a_p + \sum_{k=1}^n \gamma_k, b_p + n - \sum_{k=1}^n \gamma_k\right). \quad (\text{C10})$$

This update reflects the fraction of series currently classified as stochastic volatility processes, so that the posterior mean of p increases as more γ_k 's take the value one.

Appendix C.8. Sampling λ_3, λ_4

The sampling procedures for λ_3 (associated with the intercept) and λ_4 (associated with the impact matrix) are almost identical to those for λ_1 and λ_2 . The only differences lie in the specific subset of coefficients extracted from ϑ and the adjustment of dimensionality to match each corresponding block. Accordingly, the conditional posterior distribution of λ_3 is given by:

$$\pi(\lambda_3 \mid \vartheta, \mu, v) \propto \prod_{(l,i,j) \in S_3} v_{l,i,j}^{-\frac{1}{2}} \exp\left(-\frac{1}{2} \frac{\vartheta_{l,i,j}^2}{v_{l,i,j}}\right) \times \lambda_3^{e_{31}-1} \exp(-e_{32}\lambda_3) \quad (\text{C11})$$

$$\propto \lambda_3^{-\frac{n}{2}+e_{31}-1} \exp\left(-\frac{1}{2}\left(2e_{32}\lambda_3 + \sum_{(l,i,j) \in S_3} \vartheta_{l,i,j}^2\right)\right), \quad (\text{C12})$$

where S_3 denotes the set of coefficients governed by λ_3 . Thus,

$$\lambda_3 \mid \vartheta \sim \text{GIG}\left(-\frac{n}{2} + e_{31}, 2e_{32}, \sum_{(l,i,j) \in S_3} \vartheta_{l,i,j}^2\right). \quad (\text{C13})$$

Similarly, the conditional posterior for λ_4 that controls the shrinkage on the impact matrix coefficients can be derived as:

$$\pi(\lambda_4 \mid \vartheta, \mu, v) \propto \prod_{(l,i,j) \in S_4} v_{l,i,j}^{-\frac{1}{2}} \exp\left(-\frac{1}{2} \frac{\vartheta_{l,i,j}^2}{v_{l,i,j}}\right) \times \lambda_4^{e_{41}-1} \exp(-e_{42}\lambda_4) \quad (\text{C14})$$

$$\propto \lambda_4^{-\frac{n(n-1)}{4} + e_{41} - 1} \exp \left(-\frac{1}{2} \left(2e_{42}\lambda_4 + \sum_{(l,i,j) \in S_4} \vartheta_{l,i,j}^2 \right) \right), \quad (\text{C15})$$

where S_4 denotes the set of coefficients associated with λ_4 . Consequently,

$$\lambda_4 \mid \vartheta \sim \text{GIG} \left(-\frac{n(n-1)}{4} + e_{41}, 2e_{42}, \sum_{(l,i,j) \in S_4} \vartheta_{l,i,j}^2 \right). \quad (\text{C16})$$

Appendix D. Additional Figures and Results

Appendix D.1. RMSFE

Table D1.: Relative RMSFE of BVAR-SVVS against BVAR-SV and BVAR-CV: Medium Model

Variable	SVVS / SV				SVVS / CV			
	$h = 1$	$h = 4$	$h = 8$	$h = 16$	$h = 1$	$h = 4$	$h = 8$	$h = 16$
GDP	1.0056	1.0041	1.0122	0.9998	0.8013	0.9774*	0.9816	0.9966*
CONS	1.0038	0.9993	1.0027	1.0032	0.7510	0.9926	1.0064	0.9947*
INV	1.0869	1.0176	1.0195	1.0029	0.8464	0.9657	0.9262	0.9944
EXP	1.1567	1.0020	0.9912	1.0007	0.9622	0.9977	1.0167	1.0190
PCEX	0.9905	1.0317	1.0369	1.0004	0.9252	0.9905	0.9980	0.9966
UEMP	1.0018	1.0074	0.9725	0.9860	0.7666	0.7320	0.9072	1.0064
HOUS	0.9968	1.0492	1.0482	1.0278	0.9976	0.9315	0.9981	1.0273
SENT	0.9887	0.9386	0.9304	0.8433	0.9992	1.0018	0.9958	0.9903
TR2	0.9930	0.9997	0.9948	0.9717	0.8683**	0.9039	1.0938	1.1002
TR10	0.9931	0.9923	0.9977	1.0106	0.9172**	0.8170*	0.8657	0.9121
SP500	1.0220	1.0085	1.0370	1.0185	0.9683	0.9813	0.9823	1.0071

Note: This table reports the relative RMSFE of BVAR-SVVS against BVAR-SV and BVAR-CV across forecast horizons $h = \{1, 4, 8, 16\}$ for the medium model. Values below one indicate superior point forecast accuracy of BVAR-SVVS relative to the benchmark model. Superscripts *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively, based on the Diebold–Mariano test.

Table D2.: Relative RMSFE of BVAR-SVVS against BVAR-SV and BVAR-CV: Small Model

Variable	SVVS / SV				SVVS / CV			
	$h = 1$	$h = 4$	$h = 8$	$h = 16$	$h = 1$	$h = 4$	$h = 8$	$h = 16$
GDP	1.0003	1.0004	1.0018	1.0004	0.9990	0.9983	0.9989	0.9989
GDPD	1.0004	1.0008	1.0008	1.0014	0.9978	0.9967	0.9961	0.9945
FFR	1.0001	0.9999	0.9985	0.9993	0.9984	0.9984	0.9990	0.9989

Note: This table reports the relative RMSFE of BVAR-SVVS against BVAR-SV and BVAR-CV across forecast horizons $h = \{1, 4, 8, 16\}$ for the small model. Values below one indicate superior point forecast accuracy of BVAR-SVVS relative to the benchmark model. Superscripts *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively, based on the Diebold–Mariano test.

Table D3.: Relative RMSFE with Estimated Contemporaneous Covariance : Large Model

Variable	BVAR-SVVS / BVAR-SV				BVAR-SVVS / BVAR-CV			
	$h = 1$	$h = 4$	$h = 8$	$h = 16$	$h = 1$	$h = 4$	$h = 8$	$h = 16$
GDP	0.8131	0.9726*	0.9935	1.0260	0.8390	0.9760*	0.9893	1.0263
CONS	0.8634	0.9638*	0.9912	1.0063	0.8547*	0.9760	0.9943	1.0090
INV	0.7492	0.9461*	0.9376	1.0055	0.8114	0.9886	0.9360	1.0077
EXP	0.8836	0.9957	0.9870**	0.9930*	0.9069	1.0132	0.9879**	0.9941
IMP	0.8439	0.9671**	0.9914	1.0802	0.8739	0.9908	0.9771	1.0809
GOV	0.9912	1.0426	0.9028	1.2049	1.0268	1.0524	0.8757	1.2117
DISP	0.9955	1.0093	0.9972	1.0169	0.9556	0.9910	0.9865	1.0163
GDPP	1.0379	0.9632	0.9676*	1.1238	1.0104	0.9802	0.9982	1.1267
PCE	1.0452	0.9670**	0.9619**	1.0303	1.0066	0.9861	0.9907	1.0365
PCEX	1.0252	0.9519*	0.9383	0.9930	0.9839	0.9664	0.9729	1.0095
CPI	0.9805	0.9884	0.9483**	1.0213	0.9798	1.0061	0.9874	1.0287
POP	0.9744	0.9942	0.9935**	1.0030	0.8684*	0.9792	0.9930*	1.0047
UEMP	0.9633	0.9463	1.0122	1.0335	0.8550*	0.8779	0.9769	1.0213
WAGE	0.9791	1.0668	1.0022	1.0157	0.9573	1.0514	0.9987	1.0151
HOUS	0.9926	0.9771	0.9792*	0.9931	0.9772***	1.0034	0.9851	1.0004
IPI	0.8434	0.9848	0.9630	1.0261	0.8652	0.9965	0.9626	1.0276
CAP	0.8736	1.0363	0.9767	1.0224	0.8730	1.0427	0.9608	1.0230
SENT	0.9544	0.9895	1.0024	1.0400	0.9698	1.0293	1.0198	1.0374
TR2	0.7738	0.8552*	0.9000*	1.0654	0.8157	0.9367	0.9880	1.1308
TR5	0.9280	0.8984	0.9319	1.0404	0.9318	0.9862	1.0136	1.1004
TR10	1.0170	0.9588	0.9941	1.0406	1.0106	1.0682	1.0651	1.0997
AAA	1.1643	1.0790	1.0744	1.0838	1.1214	1.1809	1.1287	1.1257
BAA	0.9381	1.0445	1.0261	1.1080	1.0249	1.1795	1.1294	1.1478
SP500	0.9444*	0.9400	0.9524	1.0241	0.9867	0.9755	0.9733	1.0170
WTI	0.9382	1.0277	0.9976	0.9942	0.9557	1.0303	0.9976	0.9929

Note: This table reports the relative RMSFE of BVAR-SVVS against BVAR-SV and BVAR-N across forecast horizons $h = \{1, 4, 8, 16\}$. Values below one indicate superior point forecast accuracy of BVAR-SVVS relative to the benchmark model. Superscripts *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively, based on the Diebold–Mariano test.

Table D4.: Relative RMSFE with Estimated Contemporaneous Covariance : Medium Model

Variable	BVAR-SVVS / BVAR-SV				BVAR-SVVS / BVAR-CV			
	$h = 1$	$h = 4$	$h = 8$	$h = 16$	$h = 1$	$h = 4$	$h = 8$	$h = 16$
GDP	0.8875	0.9880*	1.0032	0.9999	0.8823	0.9802*	0.9957	0.9999
CONS	0.9158	0.9966	1.0025	1.0002	0.8831*	1.0062	1.0034	1.0004
INV	0.7971	0.9453**	0.9810	0.9982	0.8275	0.9305	0.9545	0.9956
EXP	0.9043	0.9844***	1.0018	1.0000	0.9097	0.9918*	0.9976	0.9998
PCEX	0.9664	1.0040	0.9995	1.0034	0.9377	1.0177	1.0024	1.0035
UEMP	1.0414	1.0638	1.0410	1.0159	0.8816	0.8618	0.9482	0.9988
HOUS	1.0123	0.9558	0.9859	0.9993	0.9862	0.9492	0.9920	1.0015
SENT	0.9933	1.0003	1.0063	0.9897	0.9964	1.0309	1.0186	1.0043
TR2	0.8445**	0.8896***	0.9552**	0.9933	0.8441*	0.9203*	0.9901	1.0048
TR10	1.0650	1.0300	1.0115	0.9799	1.0637	1.0308	1.0051	0.9856
SP500	0.9507	0.9967	1.0156	1.0036	0.9663	0.9833	1.0183	1.0027

Note: This table reports the relative RMSFE of BVAR-SVVS against BVAR-SV and BVAR-N across forecast horizons $h = \{1, 4, 8, 16\}$. Values below one indicate superior point forecast accuracy of BVAR-SVVS relative to the benchmark model. Superscripts *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively, based on the Diebold–Mariano test.

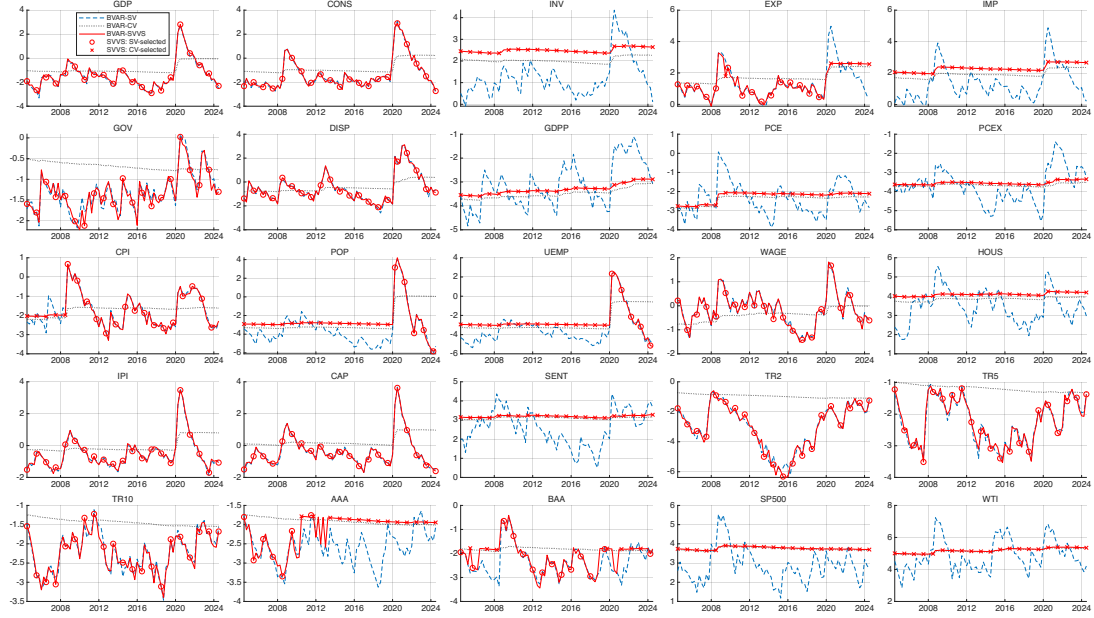
Table D5.: Relative RMSFE with Estimated Contemporaneous Covariance : Small Model

Variable	BVAR-SVVS / BVAR-SV				BVAR-SVVS / BVAR-CV			
	$h = 1$	$h = 4$	$h = 8$	$h = 16$	$h = 1$	$h = 4$	$h = 8$	$h = 16$
GDP	1.0034	0.9995	1.0000	1.0004	0.8633	0.9938	1.0004	1.0000
PCEX	0.9964	0.9984	1.0022	0.9993	0.9430*	0.9616	0.9775	0.9914
TR2	1.0001	1.0005	1.0031	1.0115	0.9282	0.9673	0.9662	0.9634***
SP500	1.0032	1.0013	1.0021	0.9999	0.9927	1.0010	1.0024	0.9993

Note: This table reports the relative RMSFE of BVAR-SVVS against BVAR-SV and BVAR-N across forecast horizons $h = \{1, 4, 8, 16\}$. Values below one indicate superior point forecast accuracy of BVAR-SVVS relative to the benchmark model. Superscripts *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively, based on the Diebold–Mariano test.

Appendix D.2. Implied Stochastic Volatility

Figure D1.: Forecast-Implied Log-Variance Paths Across Models ($h = 1$)



Note: Each Figure reports the forecast-implied log-variance path for one variable at horizon $h = 1$, comparing BVAR-SVVS with the BVAR-SV and BVAR-CV benchmarks. The red solid line denotes the forecast-implied log variance under BVAR-SVVS, the blue dashed line denotes the corresponding path under BVAR-SV, and the black dotted line denotes the corresponding path under BVAR-CV. The paths are updated recursively as the information set expands over the forecasting sample.

Figure D2.: Forecast-Implied Log-Variance Paths Across Models ($h = 4$)

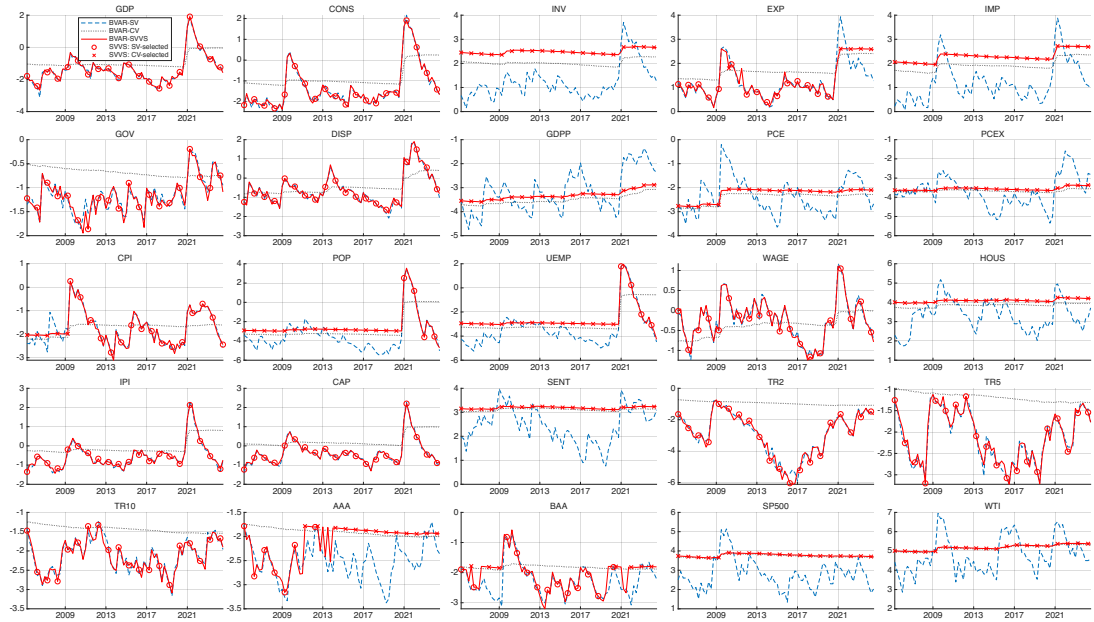


Figure D3.: Forecast-Implied Log-Variance Paths Across Models (h=8)

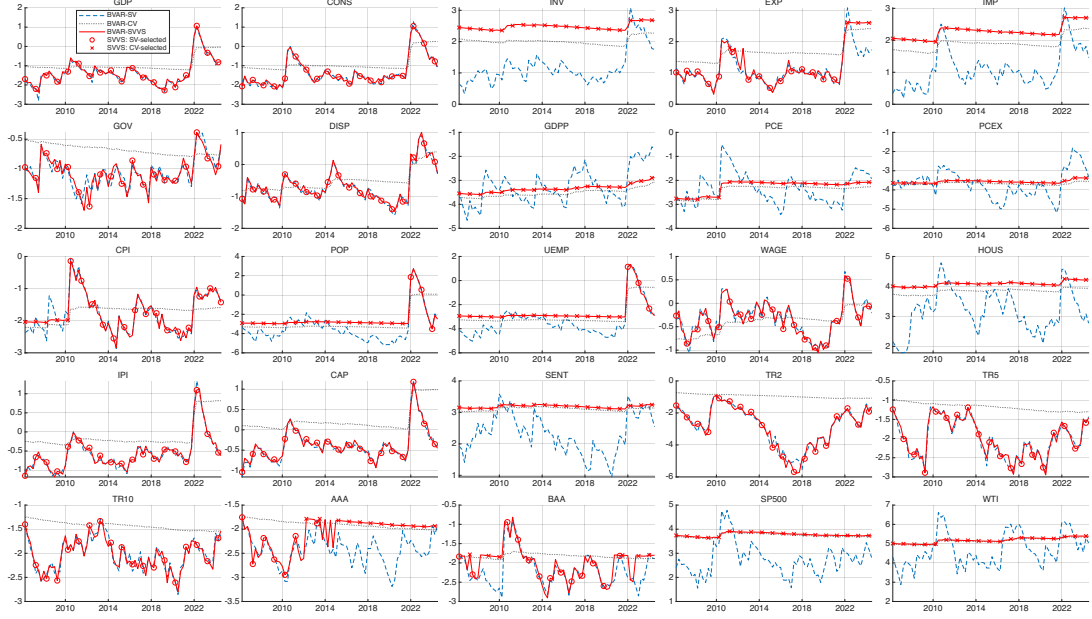
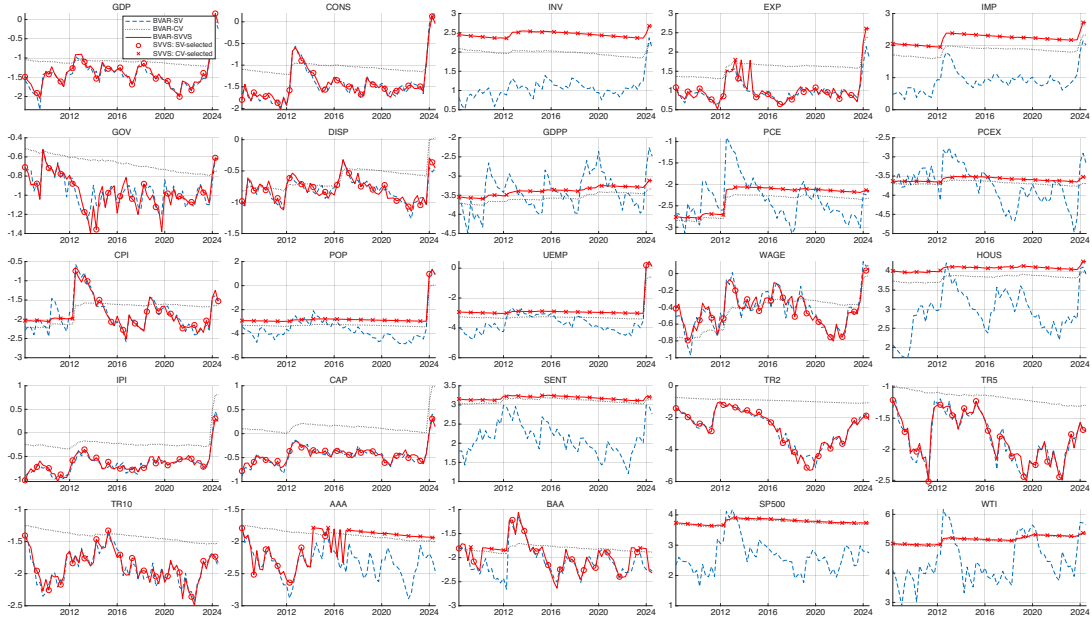


Figure D4.: Forecast-Implied Log-Variance Paths Across Models (h=16)



Appendix D.3. Decomposing Density Forecasting Gains

Appendix D.4. Density Forecasting Performance Over time

Appendix D.5. Full sample Estimation

Appendix D.6. Posterior Distributions of λ_3 and λ_4

Figure D5.: BVAR-SVVS vs. BVAR-CV ($h = 1$)

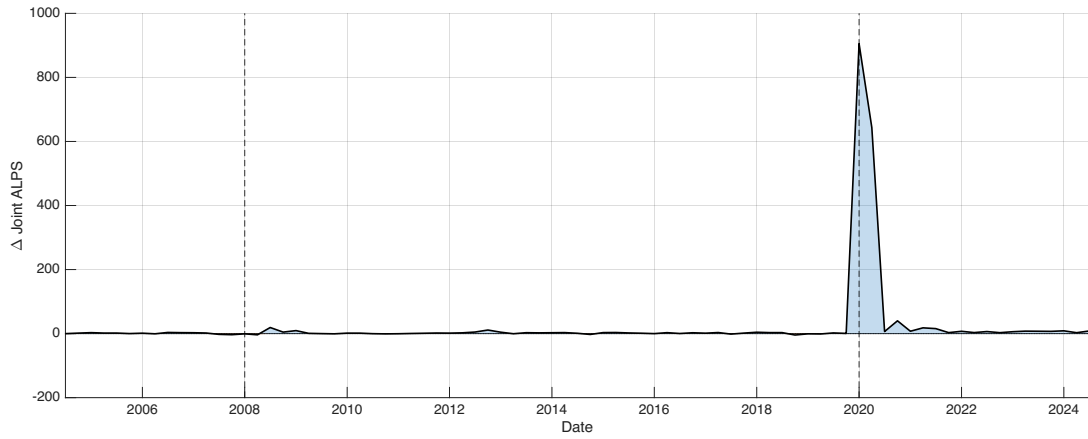


Figure D6.: BVAR-SVVS vs. BVAR-CV ($h = 8$)

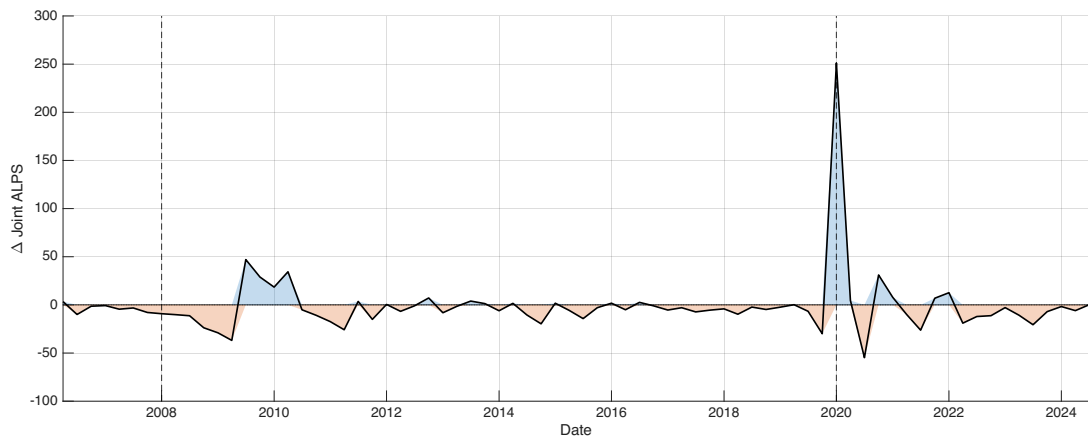


Figure D7.: BVAR-SVVS vs. BVAR-CV ($h = 16$)

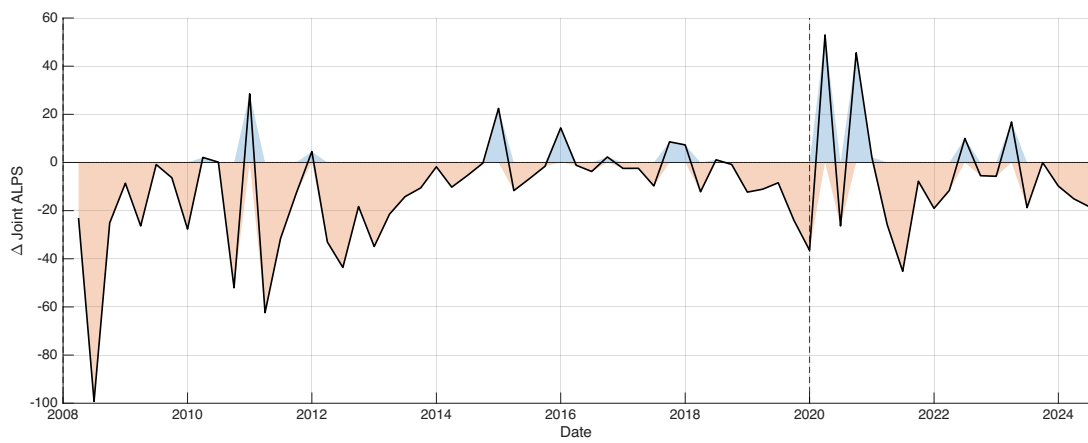


Figure D8.: BVAR-SVVS vs. BVAR-SV ($h = 1$)

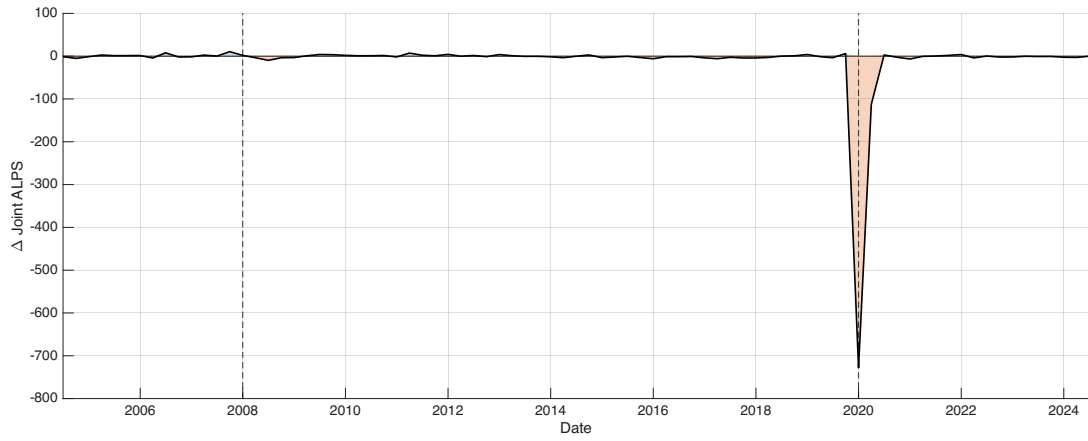


Figure D9.: BVAR-SVVS vs. BVAR-SV ($h = 4$)

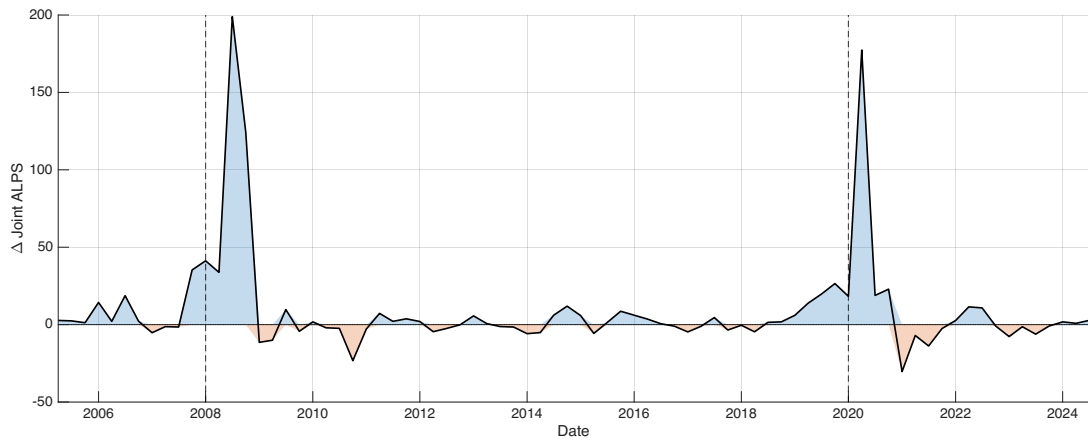


Figure D10.: BVAR-SVVS vs. BVAR-SV ($h = 8$)

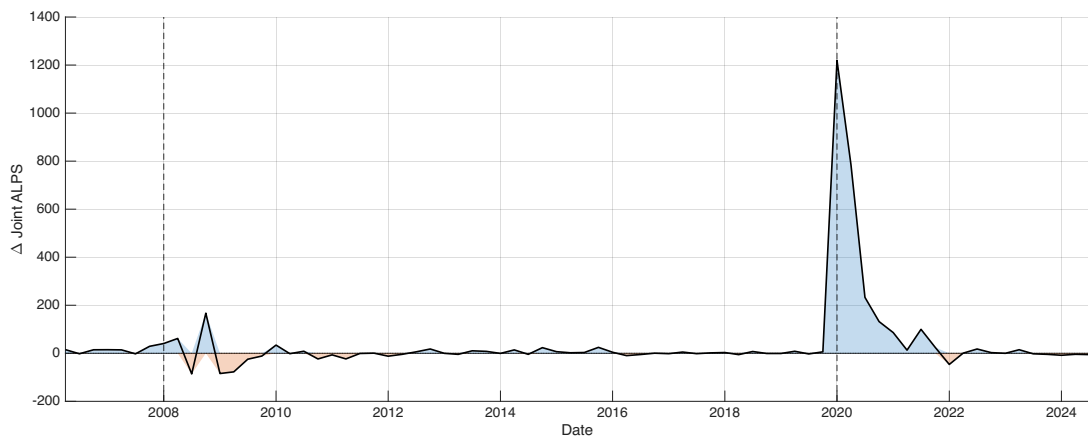


Figure D11.: Cumulative Marginal LPS: BVAR-SVVS vs. BVAR-SV ($h = 16$)

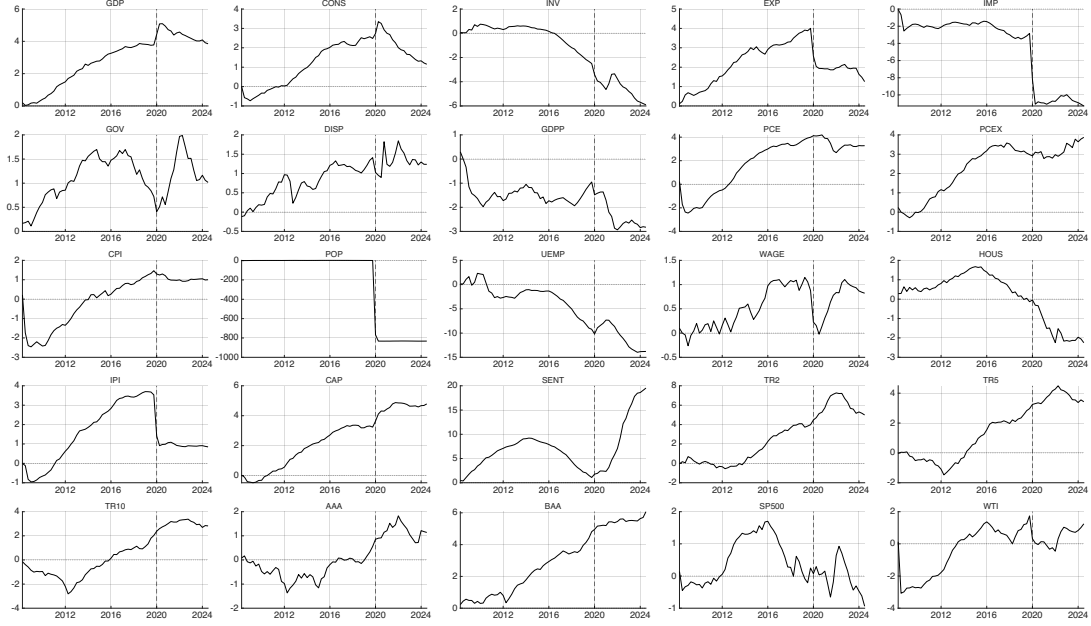
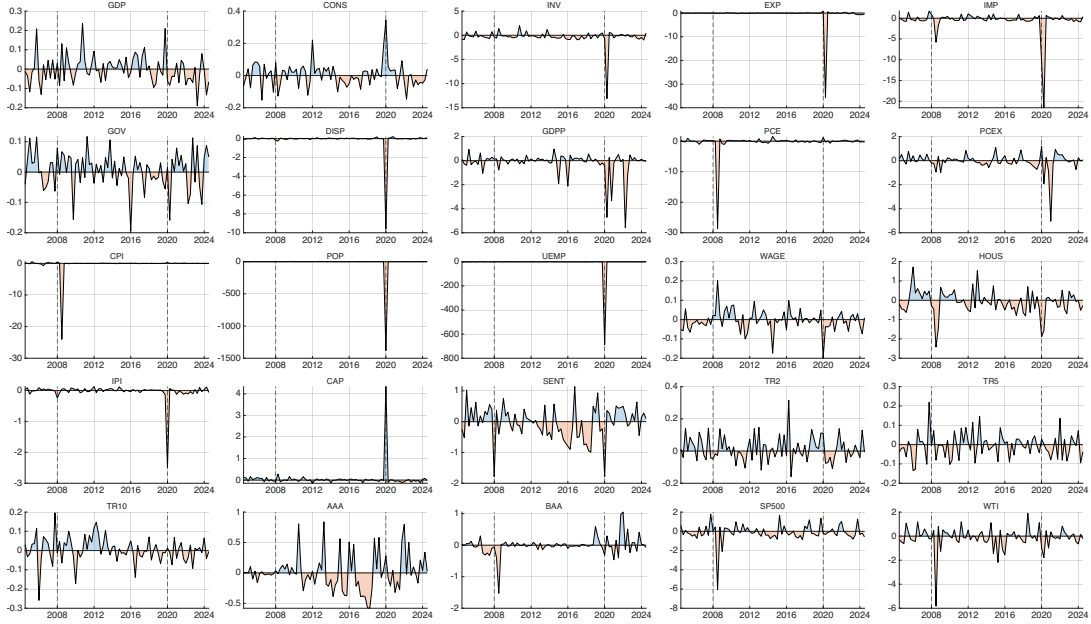


Figure D12.: Marginal LPS: BVAR-SVVS vs. BVAR-SV ($h = 1$)



Note: This figure plots the time-varying difference in Marginal Log Predictive Scores (LPS) between the BVAR-SVVS and the BVAR-SV ($h = 1$). Shaded blue areas indicate periods where the proposed model outperforms the benchmark, while shaded red areas represent underperformance. The vertical dashed lines mark the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic, respectively.

Figure D13.: Marginal LPS: BVAR-SVVS vs. BVAR-SV ($h = 4$)

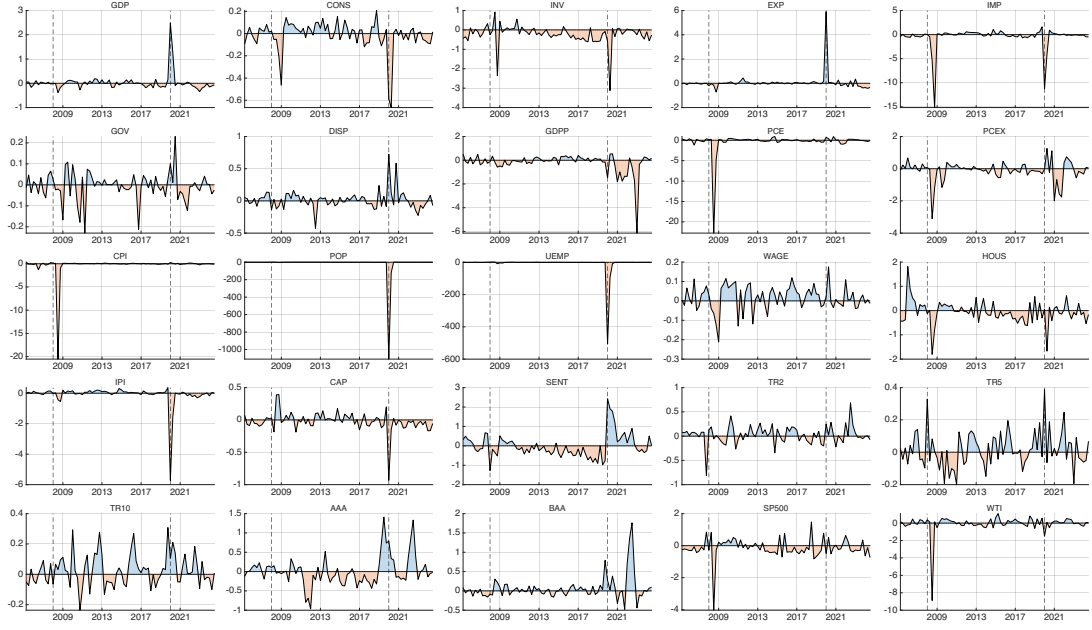


Figure D14.: Marginal LPS: BVAR-SVVS vs. BVAR-SV ($h = 8$)

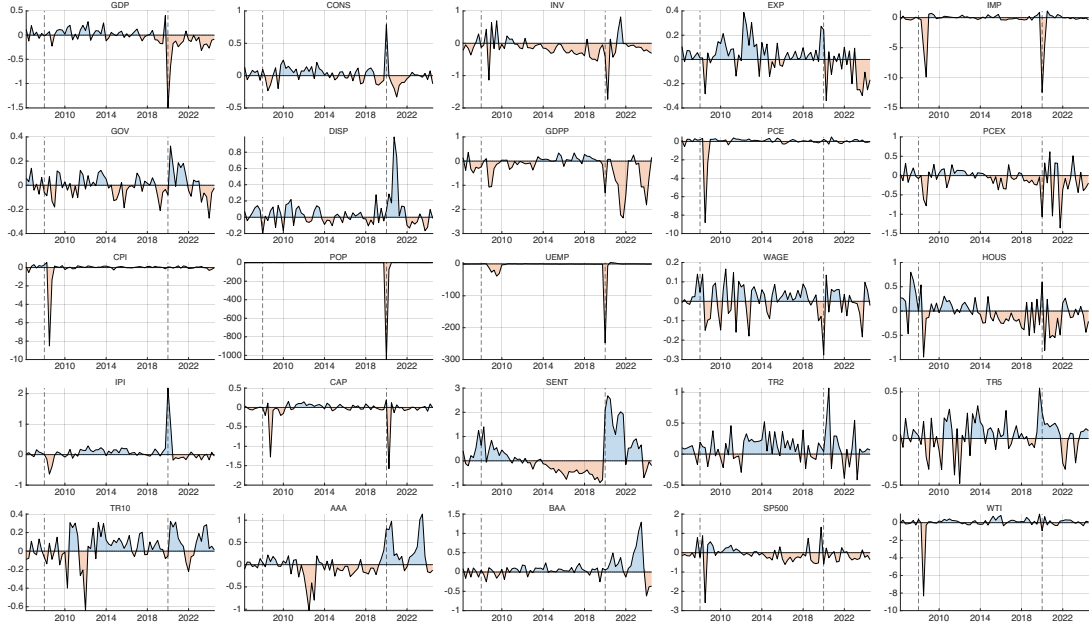


Figure D15.: Marginal LPS: BVAR-SVVS vs. BVAR-SV ($h = 16$)

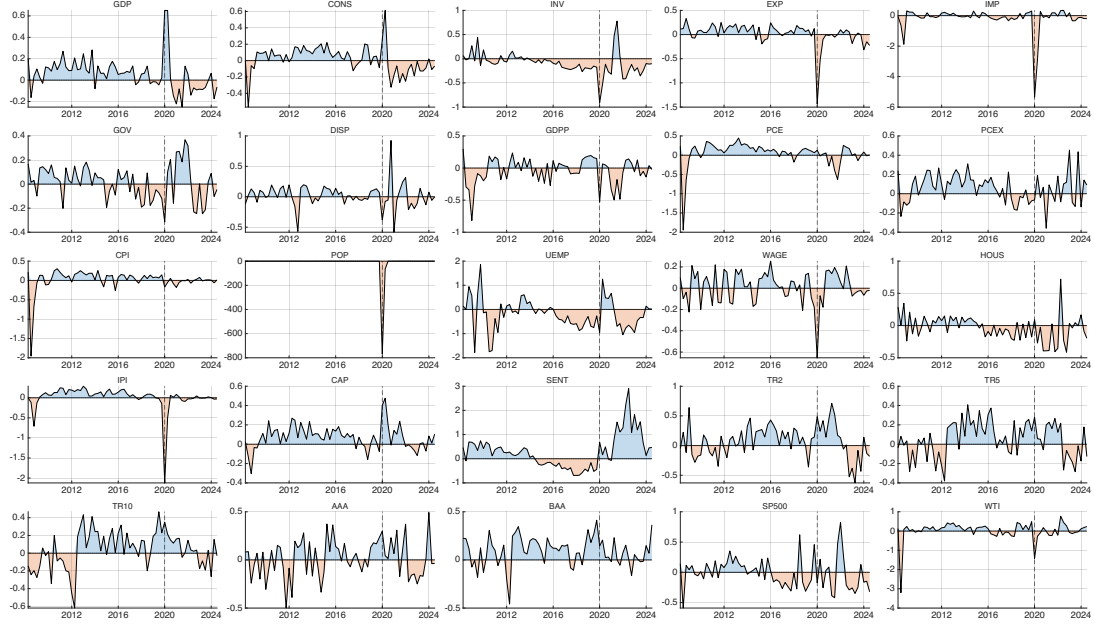


Figure D16.: Marginal LPS: BVAR-SVVS vs. BVAR-CV ($h = 1$)

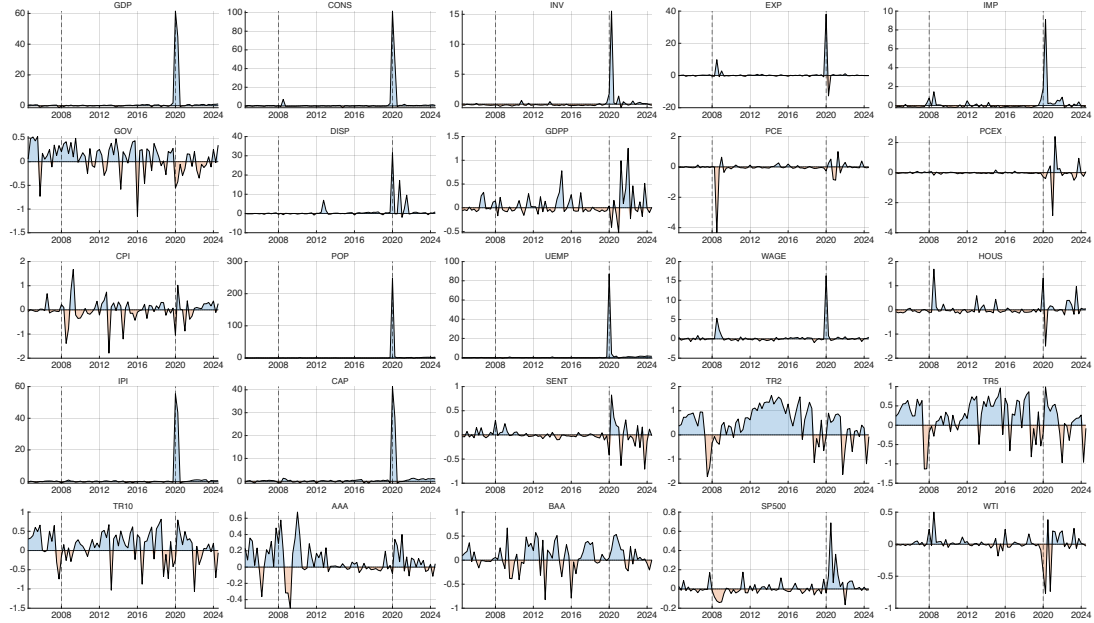
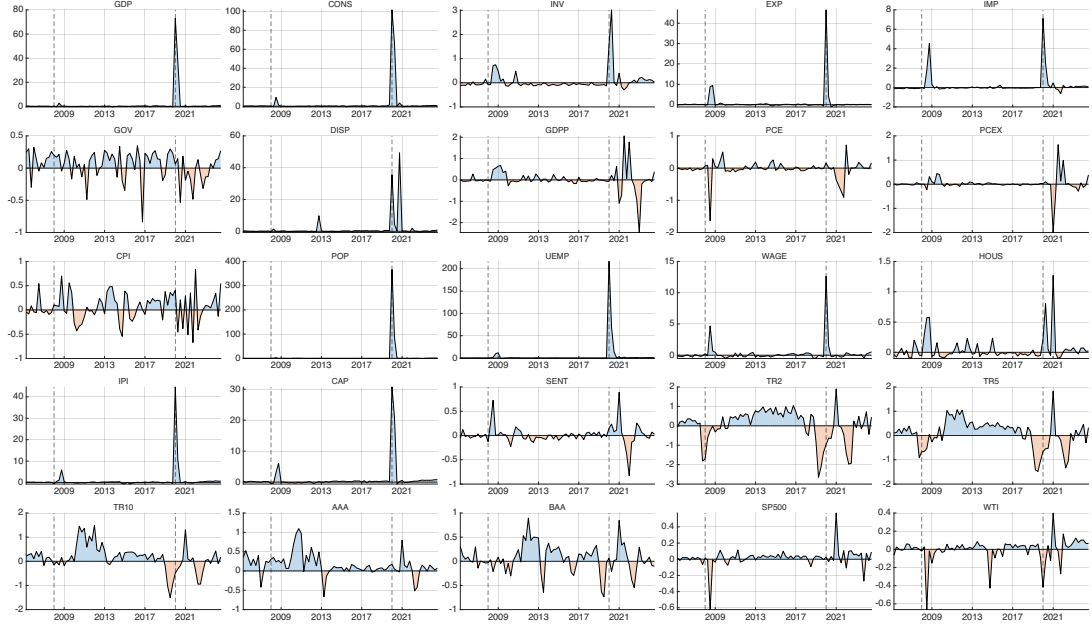


Figure D17.: Marginal LPS: BVAR-SVVS vs. BVAR-CV ($h = 4$)



Note: This figure plots the time-varying difference in Marginal Log Predictive Scores (LPS) between the BVAR-SVVS and the BVAR-CV ($h = 4$). Shaded blue areas indicate periods where the proposed model outperforms the benchmark, while shaded red areas represent underperformance. The vertical dashed lines mark the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic, respectively.

Figure D18.: Marginal LPS: BVAR-SVVS vs. BVAR-CV ($h = 8$)

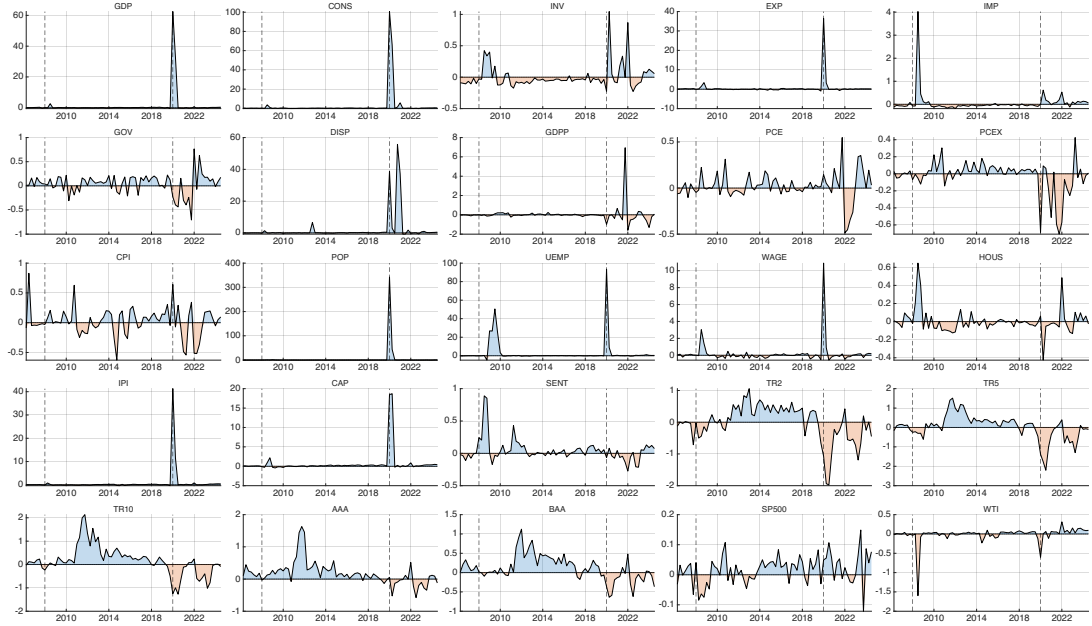


Figure D19.: Marginal LPS: BVAR-SVVS vs. BVAR-CV ($h = 16$)

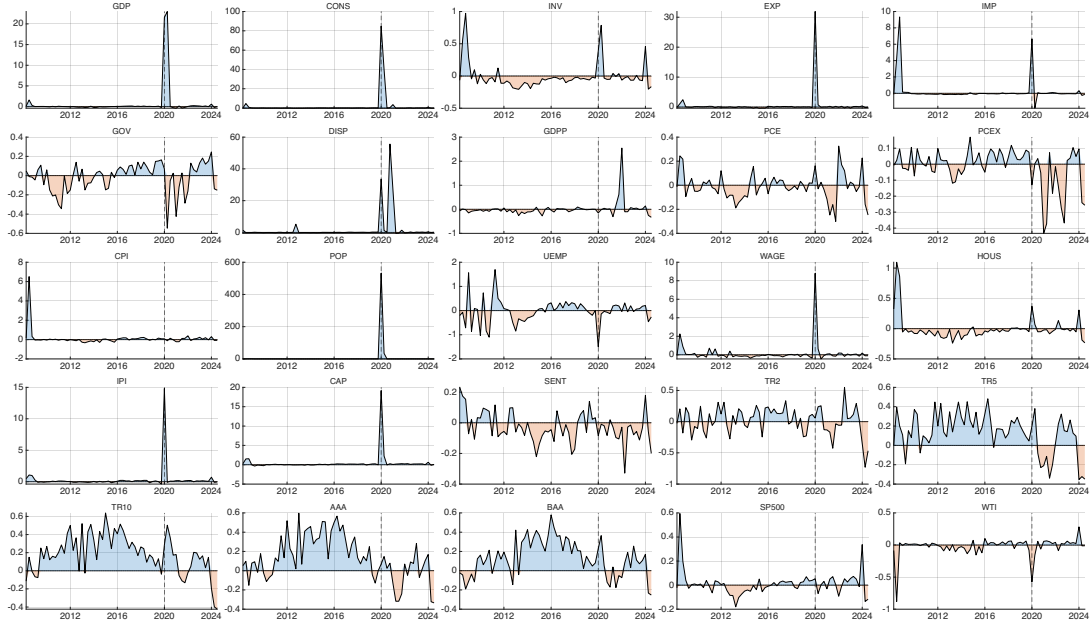


Figure D20.: Cumulative Marginal LPS: BVAR-SVVS vs. BVAR-CV ($h = 1$)

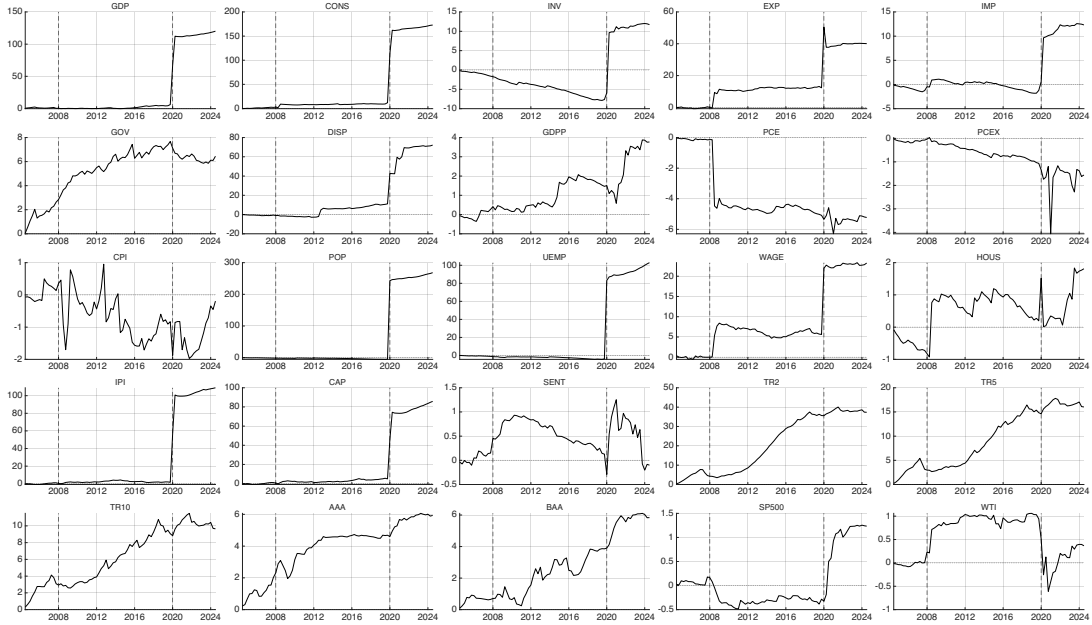


Figure D21.: Cumulative Marginal LPS: BVAR-SVVS vs. BVAR-CV ($h = 4$)

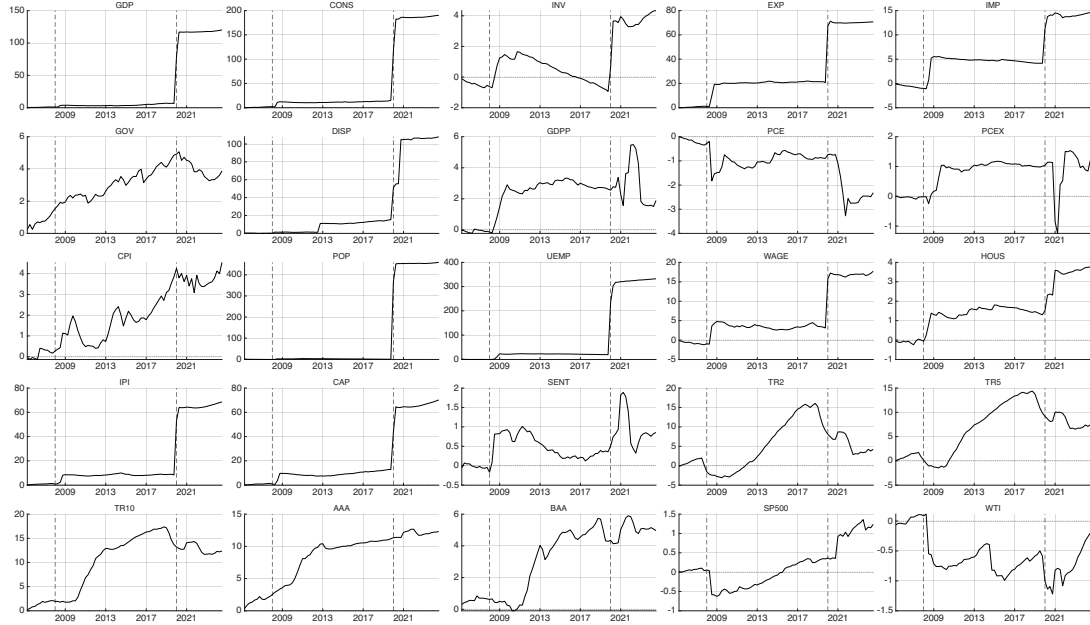


Figure D22.: Cumulative Marginal LPS: BVAR-SVVS vs. BVAR-CV ($h = 8$)

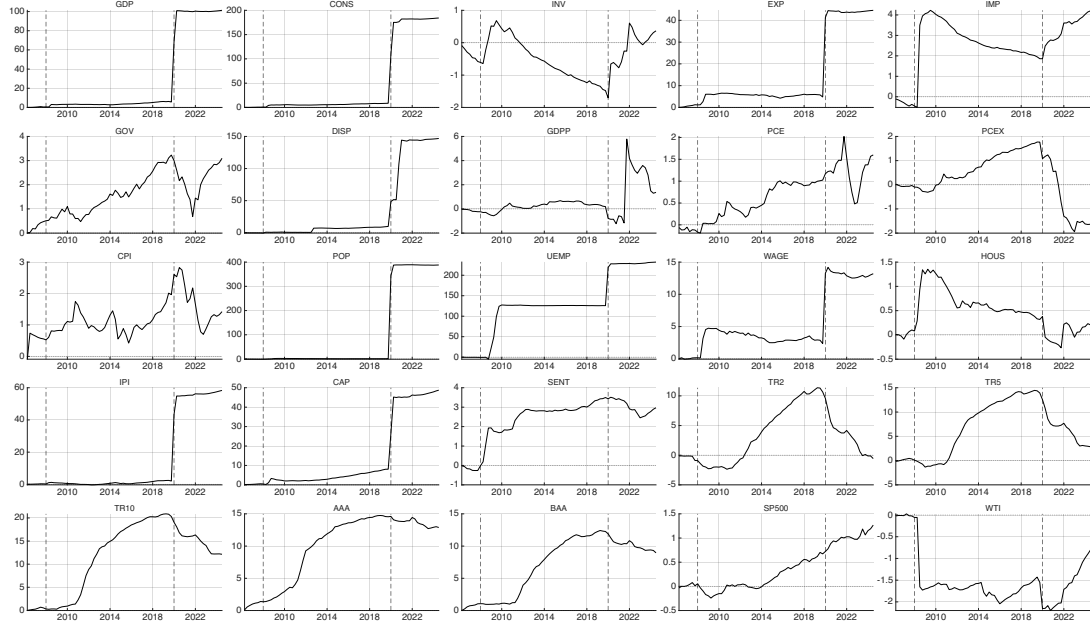


Figure D23.: Cumulative Marginal LPS: BVAR-SVVS vs. BVAR-CV ($h = 16$)

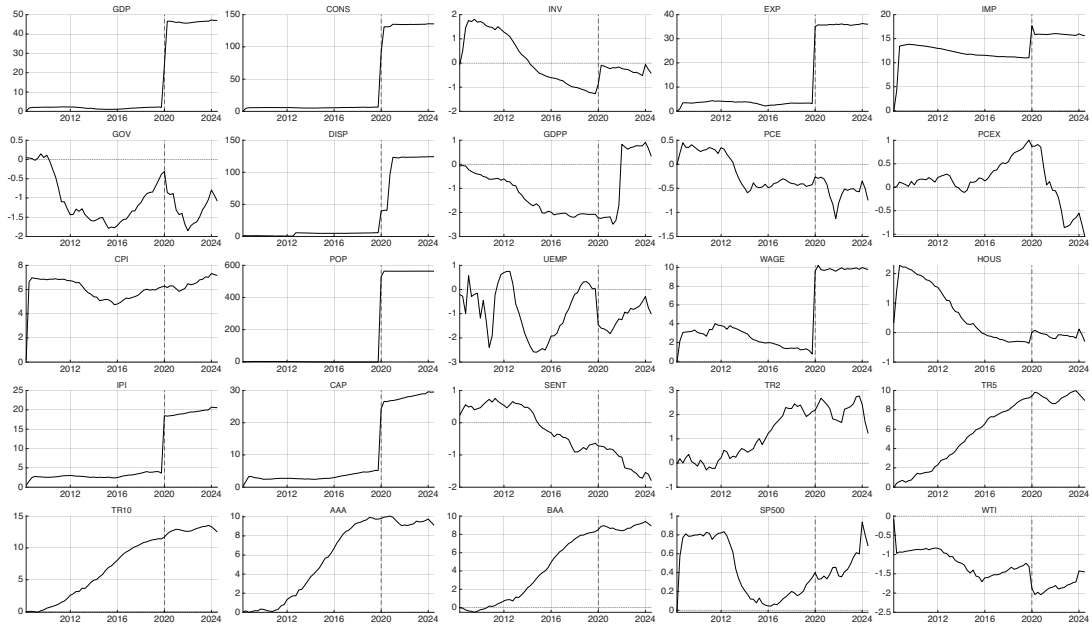


Figure D24.: Cumulative Marginal LPS: BVAR-SVVS vs. BVAR-SV ($h = 1$)

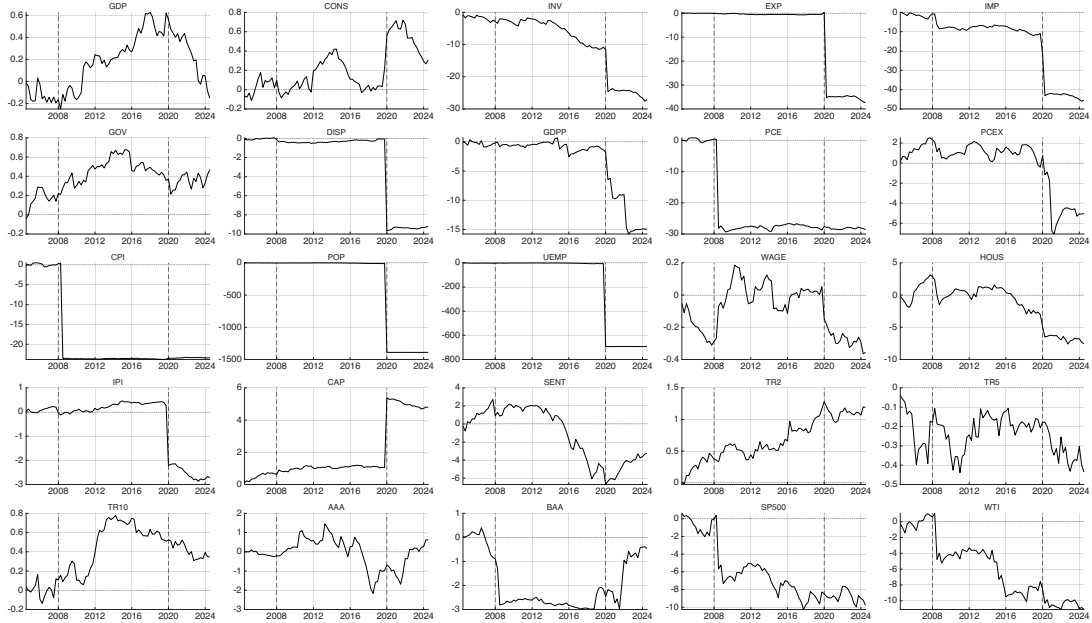


Figure D25.: Cumulative Marginal LPS: BVAR-SVVS vs. BVAR-SV ($h = 4$)

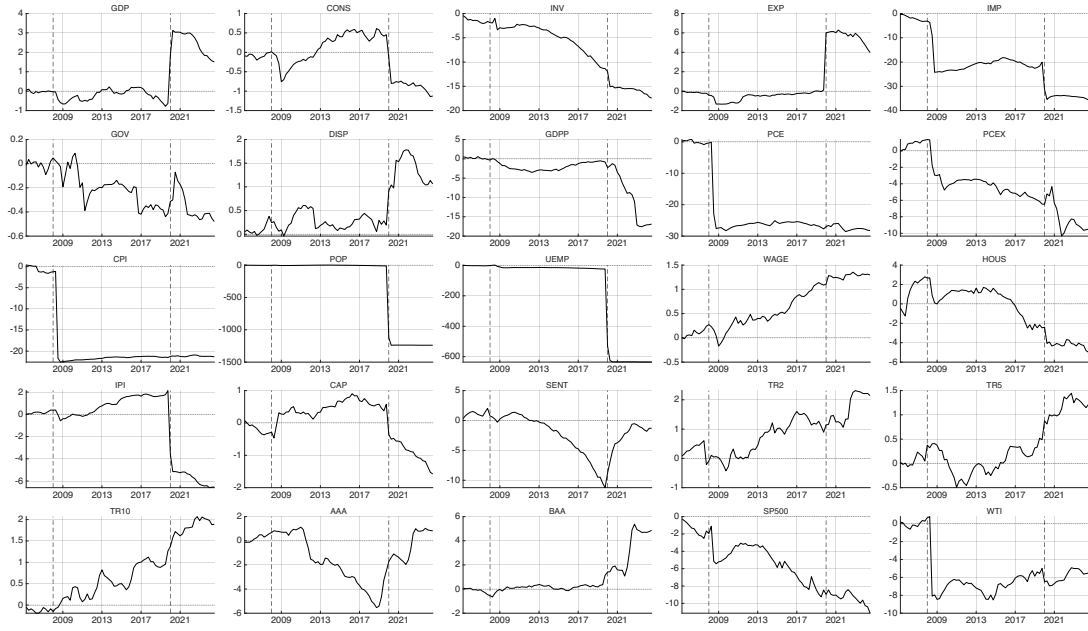


Figure D26.: Cumulative Marginal LPS: BVAR-SVVS vs. BVAR-SV ($h = 8$)

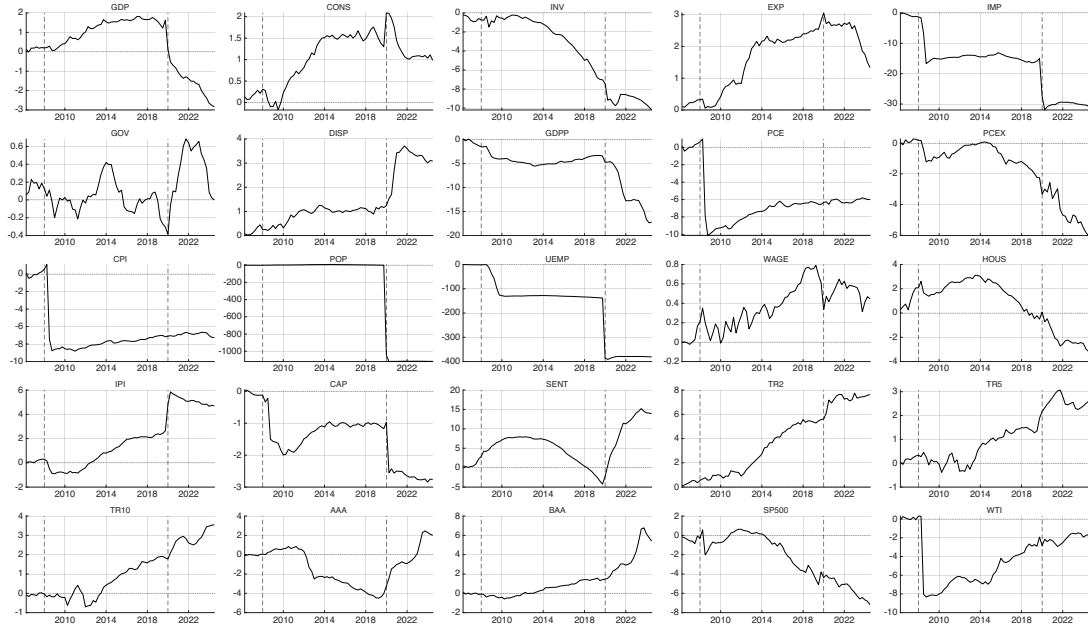
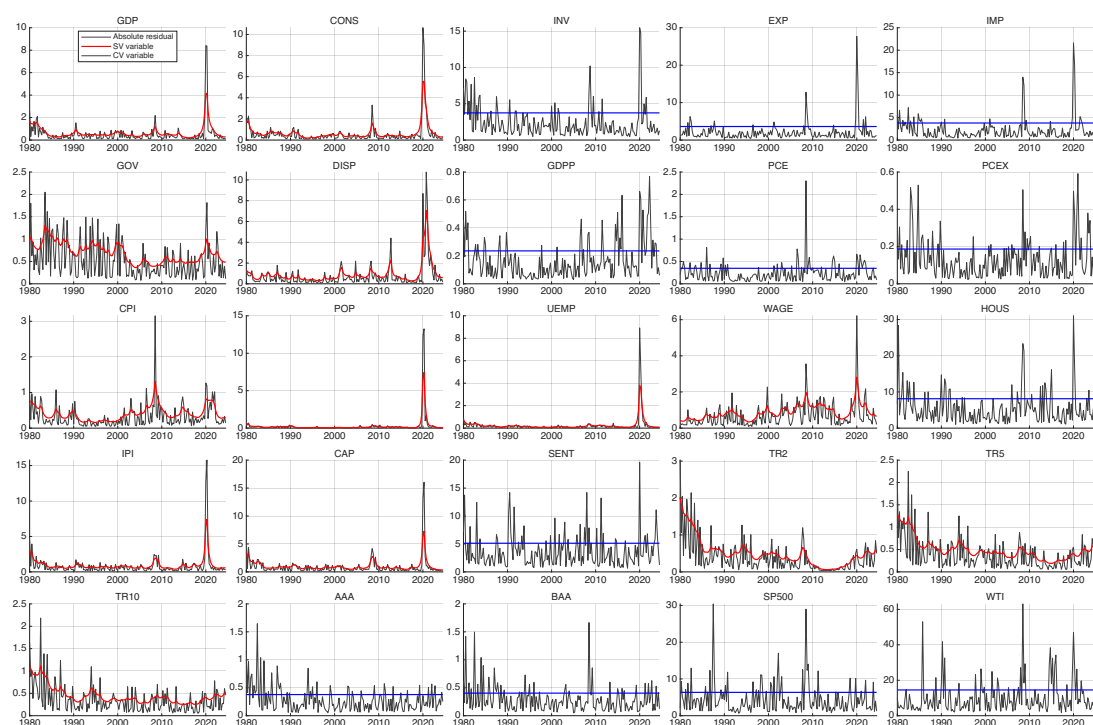
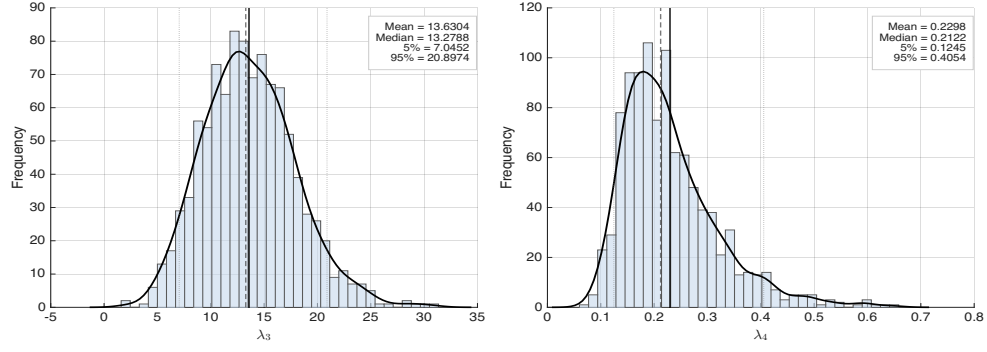


Figure D27.: Estimated Stochastic- and Constant-Volatility Specifications Across Variables

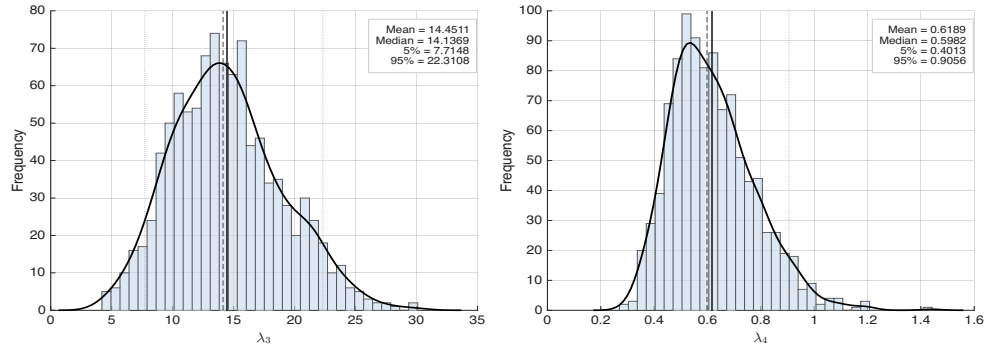


Note: Each Figure plots the absolute residuals (black line) together with the estimated volatility path. Red solid lines correspond to variables classified as stochastic-volatility processes, while blue solid lines correspond to variables classified as constant-volatility processes.

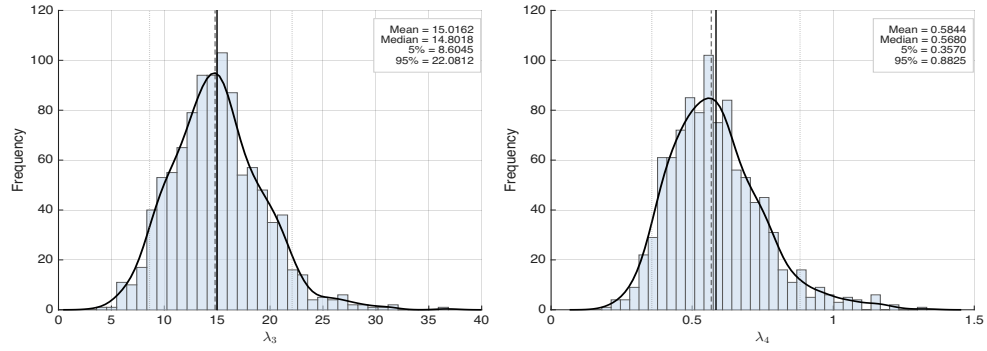
Figure D28.: Posterior Distributions of λ_3 and λ_4 in the Medium Model



(a) BVAR-SVVS

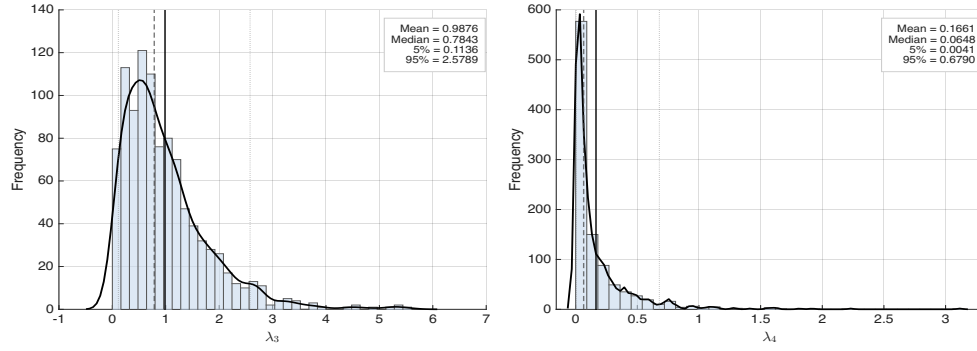


(b) BVAR-SV

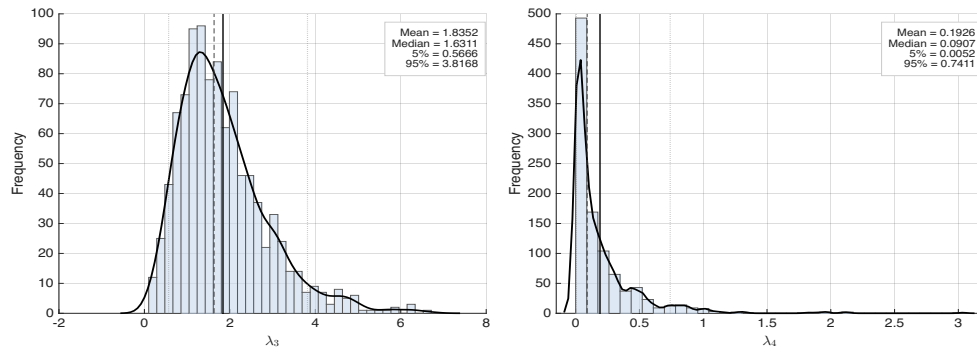


(c) BVAR-CV

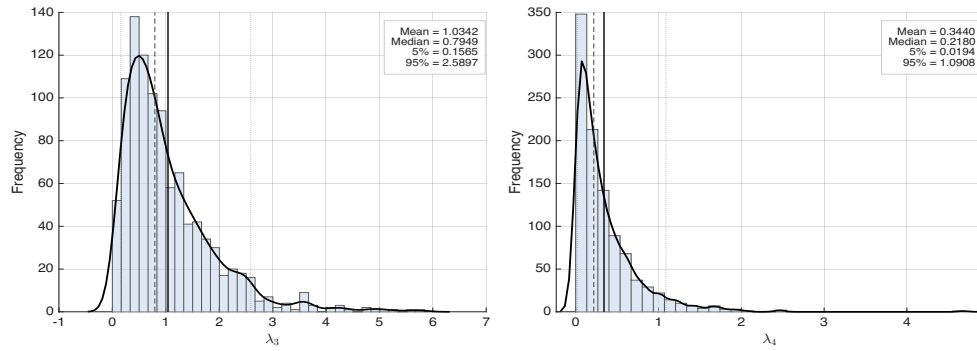
Figure D29.: Posterior Distributions of λ_3 and λ_4 in the Small Model



(a) BVAR-SVVS



(b) BVAR-SV



(c) BVAR-CV